



MAG integra

Turbomolecular Pumps with Magnetic Bearing and
Frequency Converter
Instruction manual

Operating instructions 300324726_002_C4

Part no

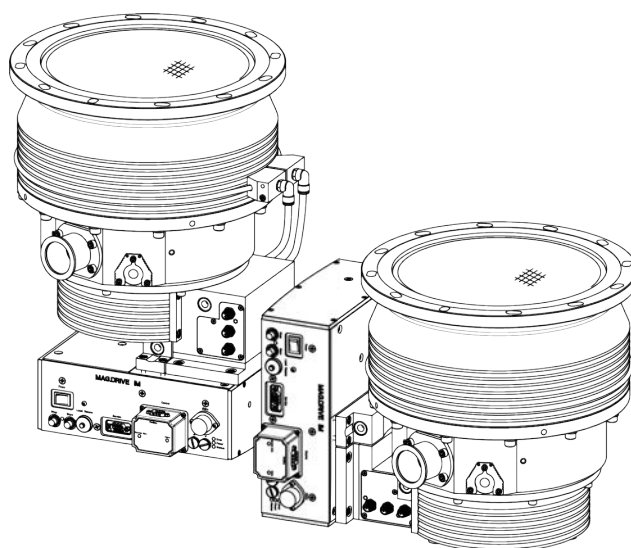
411300Vxxxx

411600Vxxxx

411700Vxxxx

412200Vxxxx

and pumps modified by
Leybold



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Associated publications

Publication title	Publication number	Link
Serial Interfaces for MAG integra	300336926	https://4vac.io/41m7a6
Ethernet/IP interface for MAG integra	300400153	https://4vac.io/rs1fqc
Profinet Interface for MAG integra	300881108	https://4vac.io/wzrcwj
MAG integra MAG W 1300 – 2200 DeviceNet	300444304	https://4vac.io/wkqrdy
MAG integra MAG W 1300 – 2200 iP/iPL EtherCat	300503445	https://4vac.io/9isspr

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We accept no liability for loss of profit, loss of market or any other indirect or consequential loss whatsoever.

Product warranty and limit of liability are dealt with in our standard terms and conditions of sale or negotiated contract under which this document is supplied.

You must use this product as described in this manual. Read the manual before you install, operate, or maintain the product. For manual enquiries, email documentation@leybold.com.

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1 Safety and compliance

For safe operation from the start, read these instructions carefully before you install or commission the equipment and keep them safe for future use. Read all the safety instructions in this section and the rest of this manual carefully and make sure that you obey these instructions.

The instruction manual is an important safety document that we often deliver digitally. It is your responsibility to keep the instruction manual available and visible while working with the equipment. Please download the digital version of the instruction manual for use on your device or print it if a device will not be available.

1.1 Definition of Warnings and Cautions

Important safety information is highlighted as warning and caution instructions which are defined as follows. Different symbols are used according to the type of hazard.

WARNING:

If you do not obey a warning, there is a risk of injury or death.

CAUTION:

If you do not obey a caution, there is a risk of minor injury, damage to equipment, related equipment or process.

NOTICE:

Information about properties or instructions for an action which, if ignored, will cause damage to the equipment.

We reserve the right to change the design and the stated data. The illustrations are not binding.

1.2 Trained personnel

For the operation of this equipment “trained personnel” are:

- skilled workers with knowledge in the fields of mechanics, electrical engineering, pollution abatement and vacuum technology and
- personnel specially trained for the operation of vacuum pumps

Safety and compliance

1.3 Safety symbols

The safety symbols on the products show the areas where care and attention is necessary.

The safety symbols that we use on the product or in the product documentation have the following meanings:

	Warning/Caution Risk of injury and/or damage to equipment. An appropriate safety instruction must be followed or a potential hazard exists.
	Warning - Automatic start up Risk of injury. The equipment can be started remotely and without warning.
	Warning - Corrosive substances Risk of injury or damage to equipment. Identifies the presence of corrosive gases, liquids or materials.
	Warning - Dangerous voltage Risk of injury. Identifies possible sources of hazardous electrical shock.
	Warning - Flammable material Risk of fire. Identifies possible sources of flammable gases, liquids or materials.
	Warning - High magnetic field Risk of injury or damage to equipment. Identifies a possible source of a magnetic field capable of disrupting equipment including pacemakers.
	Warning - Hot surfaces Risk of injury. Identifies a surface capable of inflicting burns through contact.
	Warning - Noise hazard Risk of injury. Identifies a possible source of noise above the recommended safe level.
	Warning - Overhead or Suspended load Risk of injury or damage to equipment. Identifies an area where overhead or suspended items may lower or fall.
	Warning - Risk of explosion Risk of injury or damage to equipment. Identifies a situation that could result in an explosion.
	Warning - Vacuum hazard Risk of injury. Identifies the source of a dangerous vacuum.

Safety and compliance

	Warning - Magnetic field Risk of injury. Pacemakers or other medical devices may be affected by the magnetic field.
	Mandatory action symbol Failure to comply with this action may result in injury or damage to equipment.
	Mandatory - Read the manual Failure to comply with this action may result in injury or damage to equipment.
	Warning - Use protective equipment Risk of injury. Use appropriate Personal Protective Equipment (PPE) when performing the task.

Important safety information

2 Important safety information

2.1 Mechanical hazards

1. Avoid exposing any part of the human body to the vacuum.
2. The pressure in the pump must not exceed 1.4 bar (absolute).
3. The pump is intended for generating a vacuum only. If there is a risk of an overpressure within the system and the pump, then the pump must be protected against this, by way of an overpressure safety valve, for example.
4. Vent the pump only up to atmospheric pressure.
5. When using the pump with a purge gas valve, protect the purge gas supply such that in the event of a malfunction no overpressure can occur within the system.
6. The pump must be firmly mounted to the vacuum chamber. If the mounting is not sturdy enough, pump blockage could cause the pump to break loose; internal pump components could be thrown in all directions. Never operate the pump (in bench testing, for example) without proper flanging to the vacuum chamber. Observe the information in [Attach the pump to the vacuum chamber](#) on page 28.
7. **MAG W 1300, W 1600, 1601, W 1700:** We recommend changing the rotor after 20,000 cycles at the latest.
MAG W 2200, 2201: The rotor must be changed after 10,000 cycles at the latest.
All pumps: We recommend changing the rotor after 80,000 hours of operation at the latest.
Due to high-speed and temperature, the service life of the rotor is limited. If the rotor is changed too late, it may be destroyed. Thus, in the flange mounts high forces and torque conditions can occur. The mounting screws for the pump may be torn off. When using clamped flange connections at the housing or with components above the housing, sudden twisting of the entire pump can be experienced.
8. Turbopumps as described in the following operation manual contain a high portion of kinetic energy due to their high rotational speed in combination with the specific rotor mass. In case of a malfunction of the system, for example rotor/stator contact or even a rotor crash, the rotational energy is released.
9. To avoid the destruction of the equipment and to prevent injuries of the operating staff the leading European manufacturers of vacuum pumps strictly recommend following the installation instructions as given in this manual.

2.2 Electrical hazards

1. The electrical connection must only be provided by a trained person. Please observe the national regulations in the country of use like EN 50110-1 for Europe, for example.
2. Lethal voltages are present at the mains connections. Before starting with any maintenance and service work, de-energise (lockout/tagout) the product first.
3. Unplug any connectors only when the mains voltage is switched off and the pump does no longer turn

Important safety information

4. At speeds approximately below 200 Hz, there will not be enough power any more for the LEDs, i.e. the pump may still turn with out any of the LEDs being on.
5. Unauthorized device conversion and modifications are prohibited for safety reasons.
6. Hazardous voltages are present within the frequency converter. When coming into contact with these, death or severe injury can result. After the pump has arrived at standstill, disconnect the frequency converter from the mains power and prevent it against being switched on inadvertently (lockout/tagout) before opening it. Basically there is no reason why the frequency converter should be opened. There are no user serviceable parts inside.
Should, even so, changes be made to the pump-frequency converter system then a trained electrician, as specified by the standard EN 50110-1, for example, must perform an electrical safety test again in accordance with the locally applicable safety regulations.
7. Lay connecting lines so that they cannot be damaged. Protect the lines against humidity and contact with water. Avoid any heat stress on the line due to unfavourable laying conditions.
8. Suitably support the connecting lines so that the pumps are not exposed to any major mechanical stress.
9. Do not expose pump, frequency converter and the connections to dripping water. Note the information on the IP type of protection.
10. When storing pump, frequency converter and connecting lines in a humid atmosphere, these can suffer corrosion. Corrosion gives rise to conductive deposits which in turn can cause short-circuits and reduce the insulation levels of electrical components.
11. Transport pump, frequency converter and connecting cables only in their original packaging so as to avoid any mechanical damage which in turn may reduce air gaps and creepage distances.
12. When applying external voltages above 42 V to the connection terminals, observe the applicable VDE safety regulations!
13. Make the electrical connections only after pump and accessories (e.g. air cooler) have been installed mechanically.

2.3 Thermal hazards

1. Handle the equipment only while vented and cooled down.
2. During operation of the pump certain areas can get so hot (80 °C max.) so that there is the risk of suffering burns. Protect hot parts against being touched.
3. Note the warning information on the housing surface. If these warning notices have been removed, covered or obstructed, include corresponding additional warning notices.

2.4 Hazards caused by materials and substances

1. The pump is not suited for pumping of reactive, corrosive or toxic media. If the rotor is attacked by process gases, it can suffer destruction. Thus in the flange mounts high forces and torque conditions can occur. The mounting screws for the pump may be torn off. When using clamped flange connections at the housing or with components above the housing, sudden twisting of the entire pump can be experienced.
2. The operator of the system must ensure that no plasma is to enter the pump. Otherwise it may lead to a failure of the pump.

Important safety information

3. When pumping dusty media, use a dust filter.
4. If low concentration corrosive or reactive gases are being pumped, then operate the pump with purge gas.
5. Please consult us as to which types of pump are required for specific processes and applications.
6. The forevacuum line must be tight. Hazardous gases can escape at leaks or the gases being pumped can react with air or humidity.

A leak search will always be required after having installed the pump and after service work on the vacuum.

Upon delivery the pump has an integral leak rate of $< 5 \cdot 10^{-7}$ mbar·l/s.
7. If the pump has previously handled hazardous gases, implement the proper precautionary measures before opening the intake or exhaust connection.

Before opening the pump, purge it for a longer period of time with an inert gas. If necessary, use gloves, a respirator and/or protective clothing and work under an exhaust hood. Firmly seal off the pump.

When shipping the contaminated pump for servicing, please also state the type of hazard. For this you must use a form which we have prepared for you.
8. Contaminated parts can be detrimental to health and environment.

Before beginning with any work, first find out whether any parts are contaminated. Adhere to the relevant regulations and take the necessary precautions when handling contaminated parts.

2.5 Danger of ignition

1. During operation the pressure inside the turbomolecular pump is so low that there is no danger of ignition (at pressures below about 100 mbar). A hazardous condition will be created if flammable mixtures enter the hot pump at pressures above 100 mbar. During operation the pump can reach temperatures as high as 120°C internally, and at parts of the outside surfaces 80 °C. Sparks could occur in case of damage to the pump and these could ignite explosive mixtures. Also note the safety information provided by the gas supplier.

2.6 Risk of damaging the pump

1. Never touch the rotor. Touching the rotor may cause injury and damage the rotor bearing.
2. Foreign objects which enter the pump through the intake would cause serious damage to the rotor. That's why we recommend installing an inlet screen. Damages caused during operation without the inlet screen are excluded from warranty.
3. The contact surfaces of pump housing, vacuum system and centering ring must be free of grease and dry so as to ensure sufficient stability in case the rotor seizes.
4. Connect a purge gas or venting valve to the correct flange. Confusing the forevacuum and purge gas flange can cause shock venting of the pump.
5. The interface connectors have UNC 4-40 threads. Do not use connectors with M3 threads.
6. Disconnect and connect the cable connections only while the pump is turning no longer (green status LED off) and with the mains power switched off (yellow power LED off). Otherwise there is the risk of damaging the frequency converter.

Important safety information

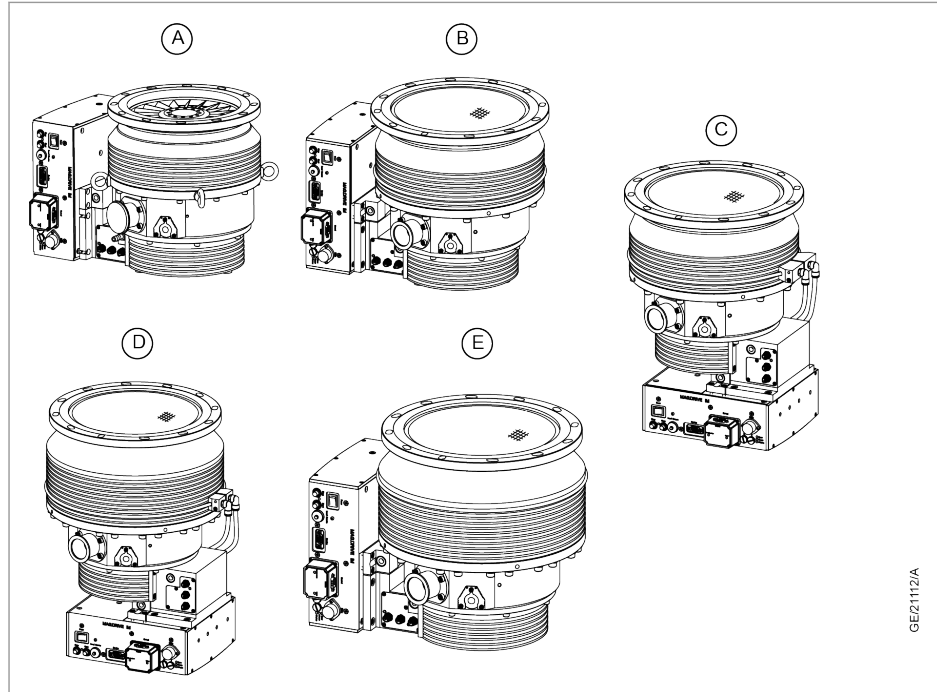
7. Do not stop the MAG with the mains. Switching off the mains while the pump is running will wear out the touch down bearings.
8. Exposure of the pump to accelerating forces must be avoided or reduced to such an extent that the rotor unit will not be excited by vibrations. In the case of critical applications you must consult our Applications Dept. first.

Connect the pump in an EMC compliant manner so as to avoid line related interference. Pressures given in bar or mbar are absolute values. If exceptionally a gauge pressure is meant, a "g" is added (bar(g)).

Description

3 Description

Figure 1. MAG integra models



- A. MAG W 1300 iP
- B. MAG W 1600 iP Booster / MAG 1601 iP Booster / MAG W 1700 iP
- C. MAG W 1600 iPL Booster / MAG 1601 iPL Booster / MAG W 1700 iPL
- D. MAG W 2200 iPL / MAG 2201 iPL
- E. MAG W 2200 iP / MAG 2201 iP

The Leybold MAG integra pumping system consists of:

The MAG integra turbo pump

The MAG are turbomolecular pumps utilizing magnetic bearings. They are designed to evacuate vacuum chambers down to pressure values in the high-vacuum range.

The required frequency converter is integrated into the pump for the MAG ... iP models.

The MAG.DRIVE electronic frequency converter

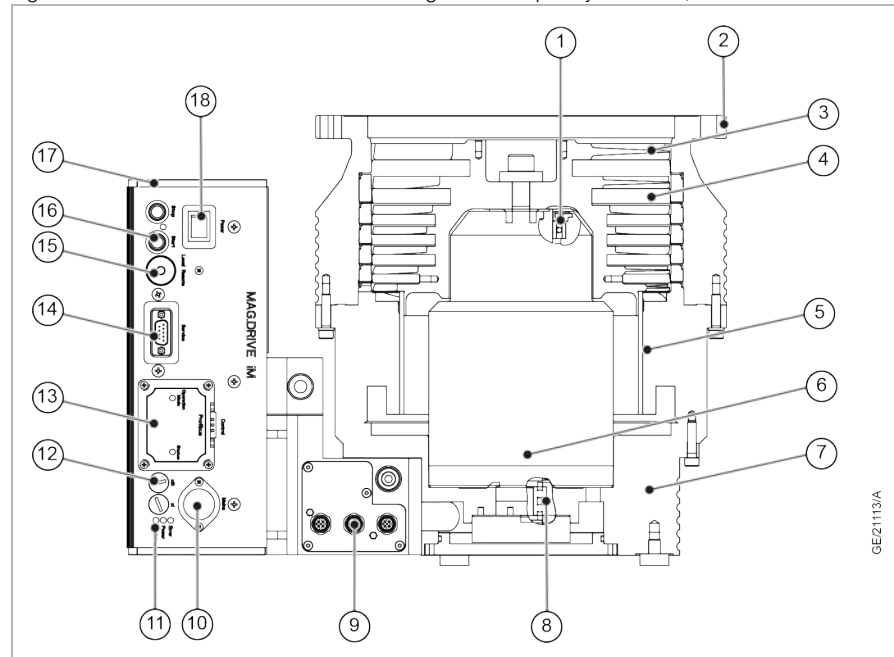
The MAG.DRIVE iM frequency converter is connected to the mains.

It outputs an AC voltage synchronised to the speed for the pump's motor.

Moreover, the electronic frequency converter processes the measured signals and controls the magnetic levitation arrangement.

Description

Figure 2. Schematic section of the MAG integra with frequency converter, other models similar



- | | |
|-------------------------------|---|
| 1. Upper touch-down bearings | 2. High-vacuum flange |
| 3. Rotor | 4. Stator elements |
| 5. Holweck stage | 6. Synchronous motor and magnetic bearing cartridge |
| 7. Base flange | 8. Bottom touch-down bearings |
| 9. 24 V DC outputs for valves | 10. Mains connection |
| 11. LEDs | 12. Address switch for Profibus |
| 13. Customer interface | 14. Service interface |
| 15. Local/Remote key | 16. Start/Stop keys |
| 17. Frequency converter | 18. On/Off switch |

3.1 Design and function

The turbomolecular pump comprises basically the pump housing, the multistage rotor with the stator package, the drive, and a magnetic bearing.

Rotor

The rotor is made from an aluminum alloy. In addition to the turbine stage the rotor has a Holweck stage.

Magnetic bearing

The MAG has a built-in precision 5-axes controlled magnetic bearing. The rotor is suspended by trouble-free magnetic bearings.

The bearing concept allows low vibration operation and insures operation of the pump in any mounting position. Magnetic bearings also guarantee ultraclean vacuum because no grease is used for lubrication of bearings.

Touch-down bearings

Touch down bearings are provided to stabilize the rotor mechanically if impacts occur during operation. They are only used in case of strong shocks or faulty electronics.

When interrupting the power supply voltage, the turbomolecular pump will operate as a generator supplying the frequency converter with power. The

Description

rotor is decelerated down to approximately 100 Hz and will then touch down on the touchdown bearings.

The MAG has a purge gas device. It is water-cooled.

A controlled synchronous motor is used to power the rotor.

Frequency converter

Drive voltage for the motor and the operating voltage for the magnetic bearing are supplied by the MAG.DRIVE iM frequency converter. It also handles the automatic monitoring of these systems.

The converter monitors continuously all important operating parameters and provides warning and alarm signals in case the operating conditions exceed the specification or the set threshold.

Switching the motor to the generator mode keeps the magnetic bearing unit in operation even in case of a mains power failure.

The frequency converter has the following interfaces:

- a Profibus or another optional interface, see [Interfaces](#) on page 45
- an interface for the Leybold Service (RS 232) and
- three connections for accessories, e.g. valves.

3.2 Supplied equipment

The pumps are shipped sealed in a PE (Polyethylene) bag with a desiccant to absorb moisture. The maximum useful life of the desiccant is one year.

The forevacuum flange is capped with a protective cap. The purge gas and venting flange is blank-flanged with centering ring with FPM (Fluororubber, resistant to temperatures up to 150°C (302 °F)) sealing ring and a clamping yoke.

The high-vacuum connection elements are not part of the standard equipment. An inlet screen has been built in.

A mains mating connector for the frequency converter side (IP 54) is supplied. A cable and a mains plug are not part of the standard equipment.

3.3 Ordering data

Table 1 Ordering data

MAG Pump with integrated converter and purge gas port, Type of protection IP 54 *		Part number	
		(Profibus)	(24 V PLC Interface)
MAG W 1300 iP	DN 200 ISO-F	411300V0504	411300V0514
MAG W 1300 iP	DN 200 CF	411300V0506	411300V0516
MAG W 1300 iPL	DN 200 ISO-F	411300V0704	411300V0714
MAG W 1300 iPL	DN 200 CF	411300V0706	411300V0716
MAG W 1600 iP Booster	DN 250 ISO-F	411600V0504	411600V0514
MAG W 1600 iPL Booster	DN 250 ISO-F	411600V0704	411600V0714
MAG 1601 iP Booster	DN 250 ISO-F	411600V2504	411600V2514
MAG 1601 iPL Booster	DN 250 ISO-F	411600V2704	411600V2714
MAG W 1700 iP	DN 250 ISO-F	411700V0504	411700V0514
MAG W 1700 iP	DN 250 CF	411700V0506	411700V0516

Description

MAG Pump with integrated converter and purge gas port, Type of protection IP 54 *		Part number	
		(Profibus)	(24 V PLC Interface)
MAG W 1700 iPL	DN 250 ISO-F	411700V0704	411700V0714
MAG W 1700 iPL	DN 250 CF	411700V0706	411700V0716
MAG W 2200 iP	DN 250 ISO-F	412200V0504	412200V0514
MAG W 2200 iP	DN 250 CF	412200V0506	412200V0516
MAG W 2200 iPL	DN 250 ISO-F	412200V0704	412200V0714
MAG W 2200 iPL	DN 250 CF	412200V0706	412200V0716
MAG 2201 iP	DN250 ISO-F	412200V2504	412200V2514
MAG 2201 iPL	DN 250 ISO-F	412200V2704	412200V2714

other interfaces on request

* The IP 54 type of protection is only ensured with an IP 54 protection cap on the customer interface.

Pump versions without IP 54 protection cap can only be classified as IP 20. Please consult with us for an optional IP 54 protection cap.

Technical data

4 Technical data

Table 2 Technical data

Parameter	Unit	W1300 iP(L)	W1600 iP(L) Booster	1601 iP(L) Booster	W 1700 iP(L)	W 2200 iP(L)	2201 iP(L)
High vacuum flange	DN	250 ISO-F 200 CF	250 ISO-F	250 ISO-F	250 ISO-F 250 CF	250 ISO-F 250 CF	250 ISO-F
Pumping speed (PNEUROP)							
▪ N ₂	l s ⁻¹	1100	1600	1600	1610	2100	2100
▪ Ar		1050	1470	1470	1480	1900	1900
▪ He		1220	1770	1700	1710	2050	2050
▪ H2		1130	1570	1500	1660	1750	1750
Maximum compression							
▪ N ₂		> 10 ⁸	> 10 ⁷	> 10 ⁷	> 10 ⁸	> 10 ⁸	> 10 ⁸
▪ Ar		> 10 ⁸	> 10 ⁷	> 10 ⁷	> 10 ⁸	> 10 ⁸	> 10 ⁸
▪ He at 1 sccm		2·10 ⁵	6·10 ⁴	3·10 ³	2·10 ⁵	5·10 ⁴	5·10 ³
▪ H ₂ at 1 sccm		8·10 ³	1·10 ³	5·10 ²	4·10 ³	5·10 ³	5·10 ²
Maximum gas throughput *#							
▪ N ₂ temporary, for pump down	mbar l s ⁻¹	30	60	60	30	30	50
▪ N ₂ continuous operation		20	30	40	20	17	36
▪ Ar temporary, for pump down		20	30	30	20	20	30
▪ Ar continuous operation		15	20	25	15	12	24
Maximum forevacuum pressure # N ₂ /Ar	mbar	4	1	1	4	2.5	1.2
Nominal speed	min ⁻¹	37,800	33,000	33,000	33,000	30,6000	30,000
Stand-by speed to be set between	min ⁻¹	13,800 (230 Hz) and nominal speed					
Ultimate pressure ISO-F / CF flange	mbar	< 10 ⁻⁸ / < 10 ⁻¹⁰					
Maximum bake-out temperature	°C	80					
Run-up time	min	< 5 min	< 7 min	< 7 min	< 7 min	< 10 min	< 10 min
Forevacuum flange	DN	40 KF					
Purge gas / venting flange	DN	16 KF (clamp shoe)					
Cooling connection		G 1/8"					
Weight, approximate	kg	40	45	45	45	50	50

Technical data

Parameter	Unit	W1300 iP(L)	W1600 iP(L) Booster	1601 iP(L) Booster	W 1700 iP(L)	W 2200 iP(L)	2201 iP(L)
Noise level acc. to ISO 3744	dB (A)	< 41					
Vibration level at high-vacuum flange at max. speed	µm	< 0.01					

* The maximum gas throughput for continuous operation in connection with N₂ or Ar is attainable with a cooling water temperature below 25 °C and a forevacuum pump system having a pumping speed of over 600 m³/h. These values may differ for other operating conditions.

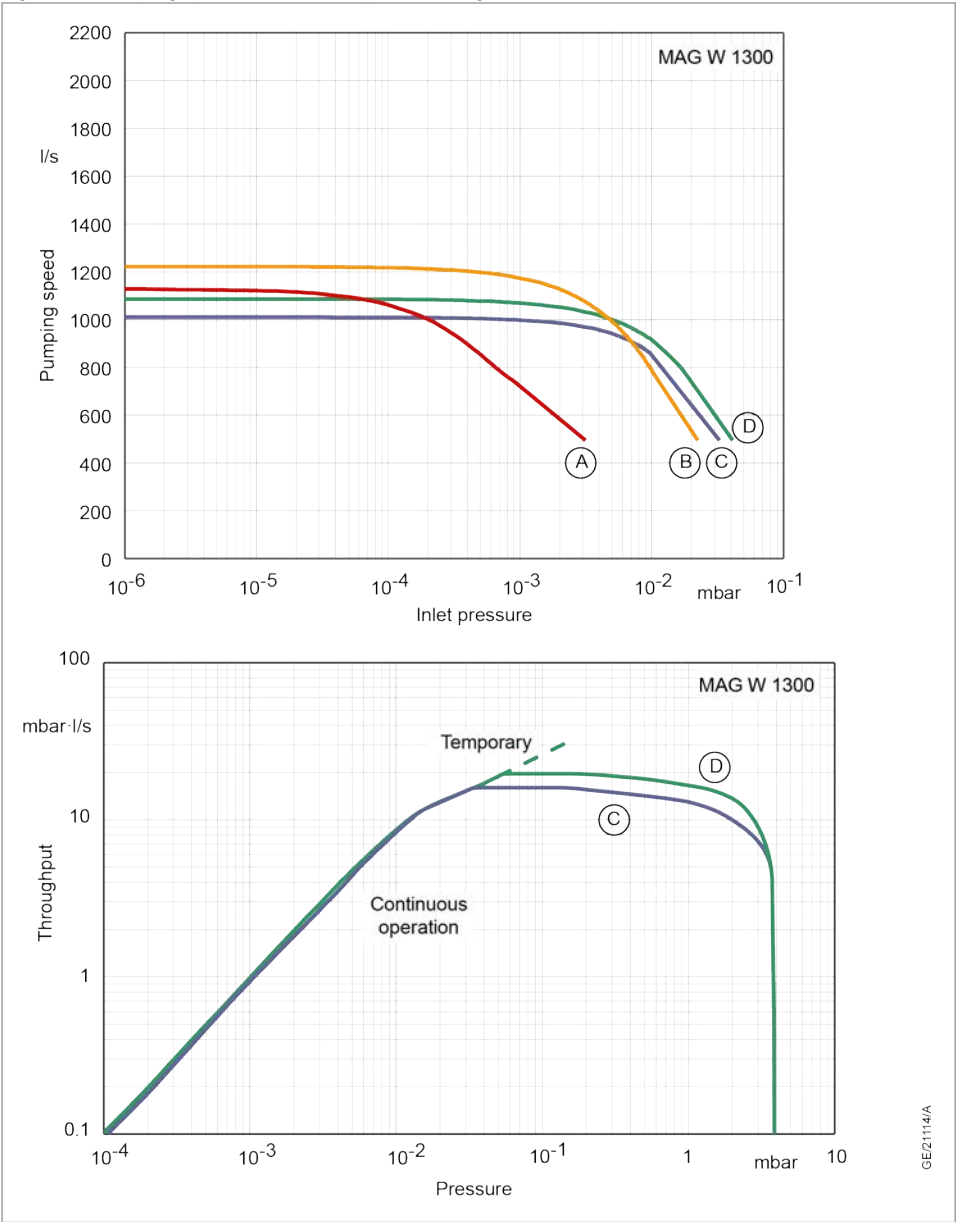
Maximum gas throughput and maximum forevacuum pressure cannot be attained simultaneously.

Table 3. Technical data for the integrated frequency converter

Parameter	Value
Supply voltage	200 to 240 ± 10% V a.c.
Power consumption maximum at ultimate pressure	750 VA 150 VA
Mains frequency	50/60 Hz
Max. load capability of the relays (interface X1, optional)(for SELV or PELV power circuits only)	32 V, 0.5 A
Max. load for each valve output	24 VDC, 0.35 A
Temperature during operation	10 to 45 °C
Storage temperature	–10 to 60 °C
Relative air humidity (non condensing)	5 to 85 %
Type of protection	IP 54
Overvoltage category	II
Contamination level acc. to IEC 61010	2

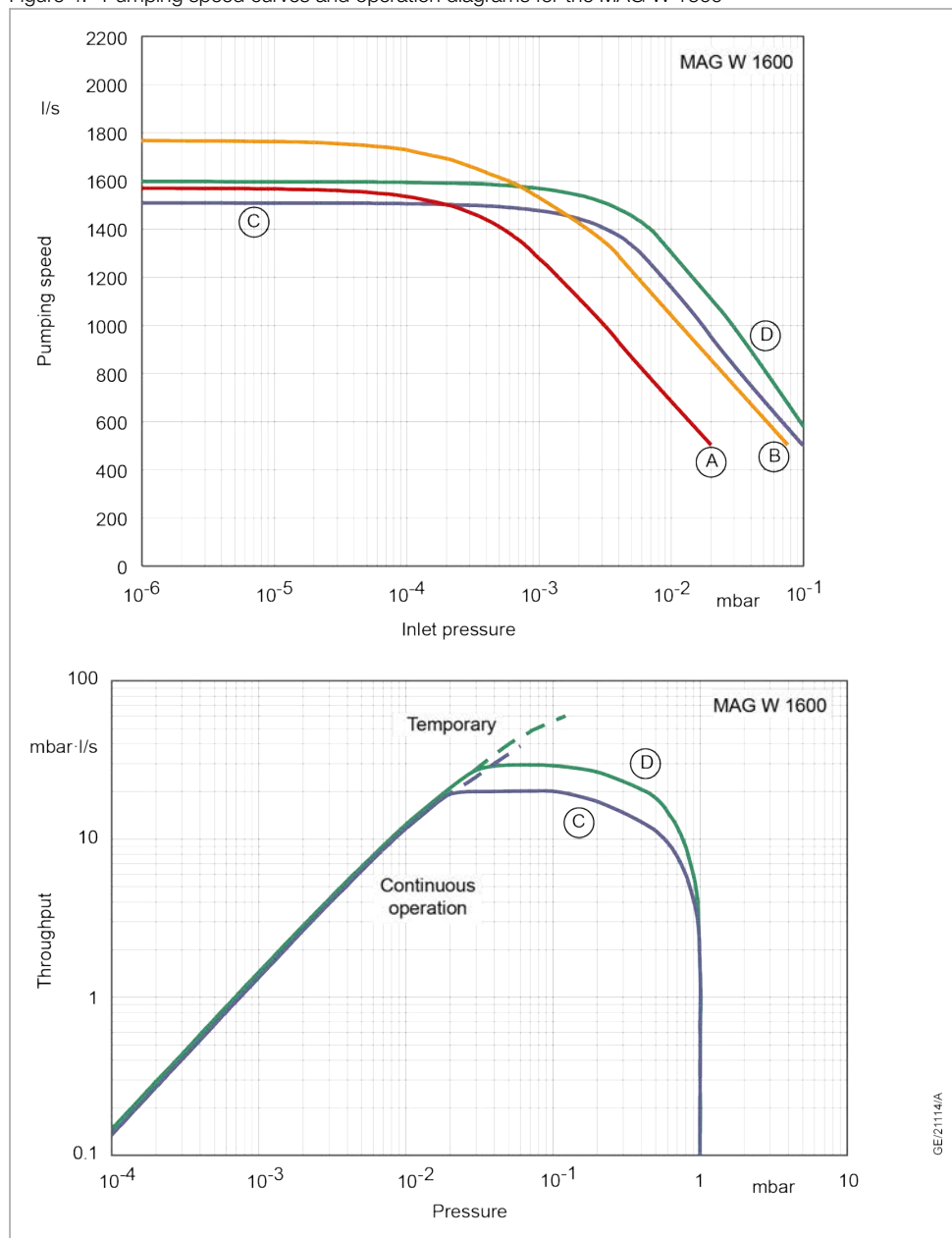
Technical data

Figure 3. Pumping speed curves and operation diagrams for the MAG W 1300



Technical data

Figure 4. Pumping speed curves and operation diagrams for the MAG W 1600

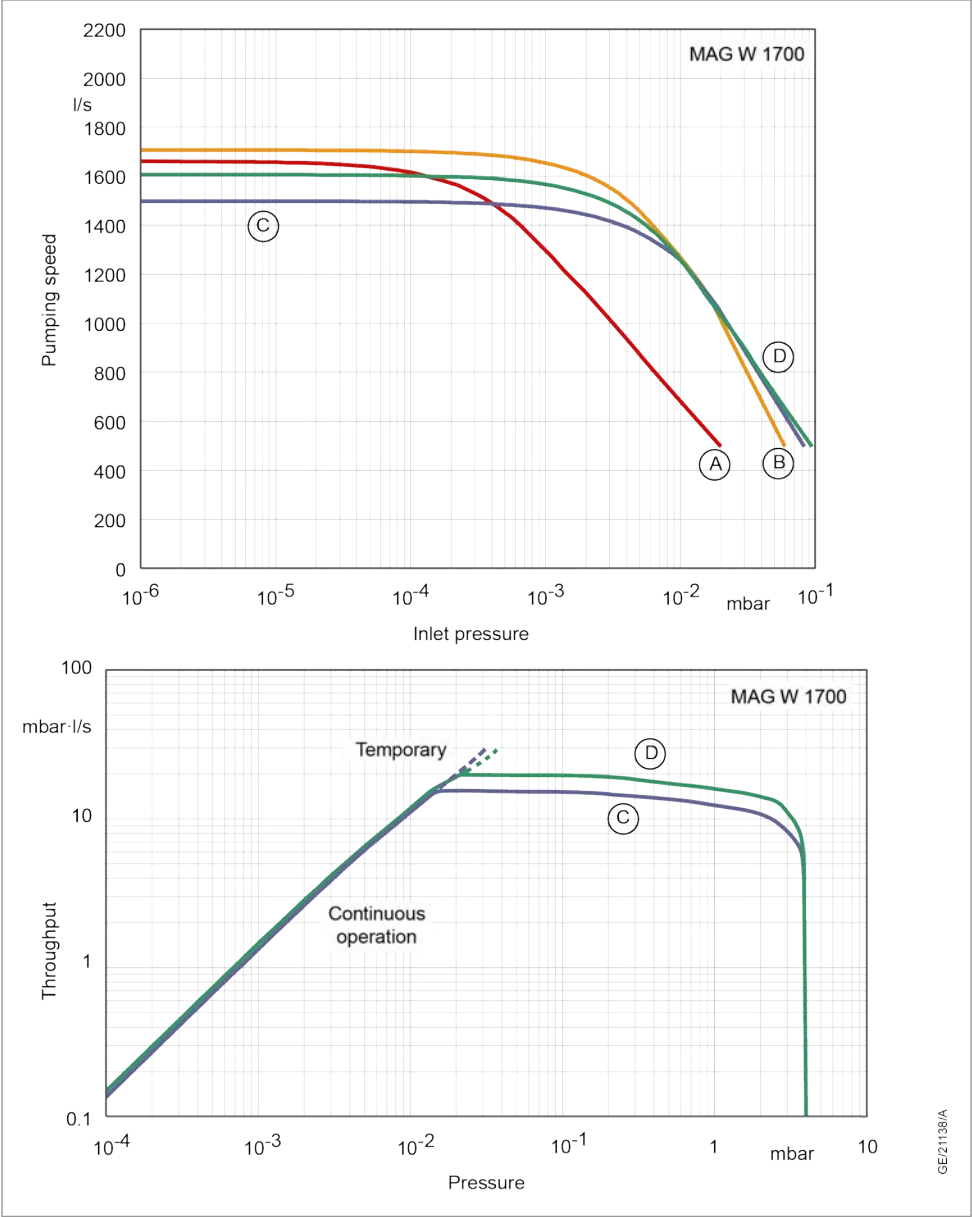


A. Hydrogen
C. Argon

B. Helium
D. Nitrogen

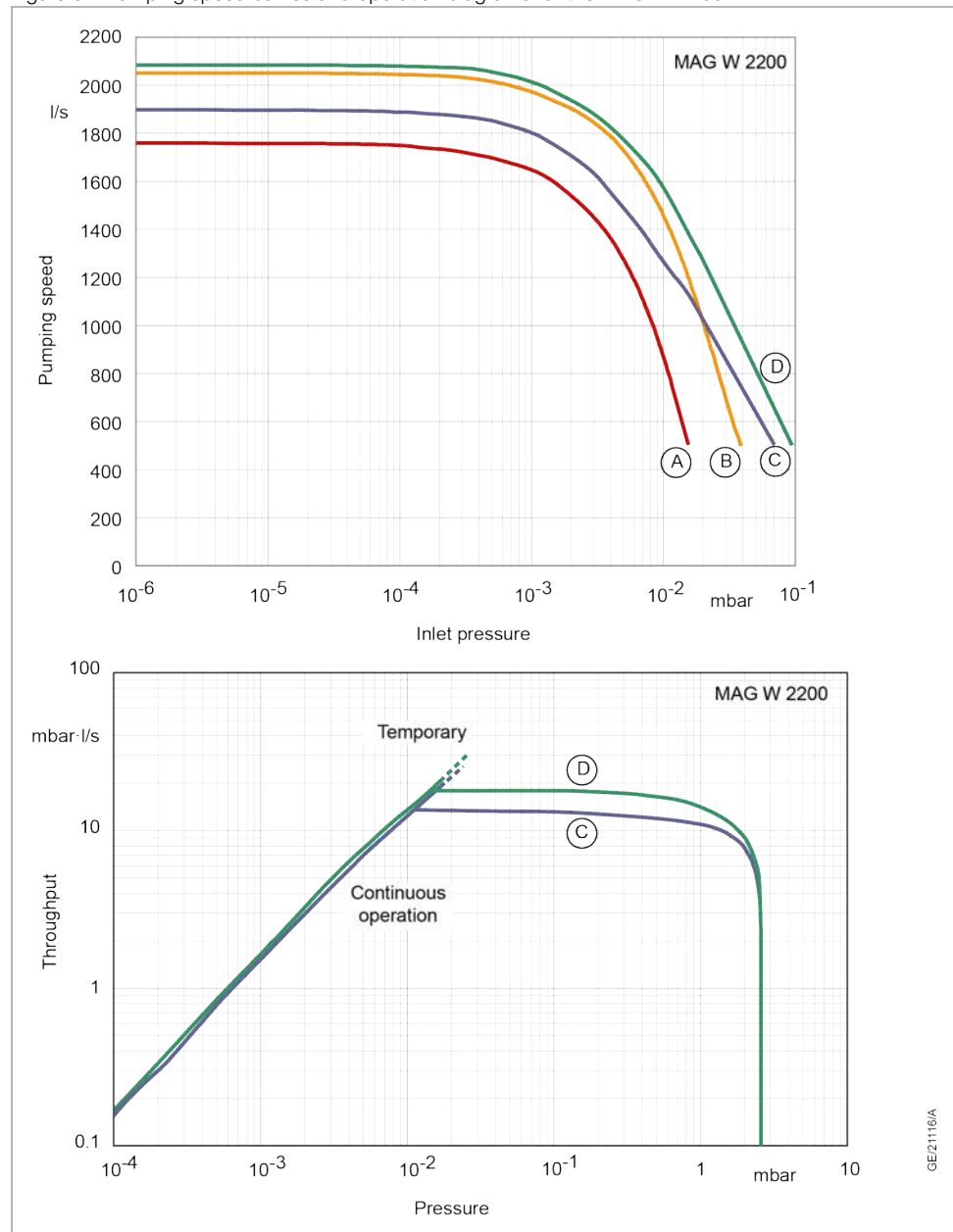
Technical data

Figure 5. Pumping speed curves and operation diagrams for the MAG W 1700



Technical data

Figure 6. Pumping speed curves and operation diagrams for the MAG W 2200

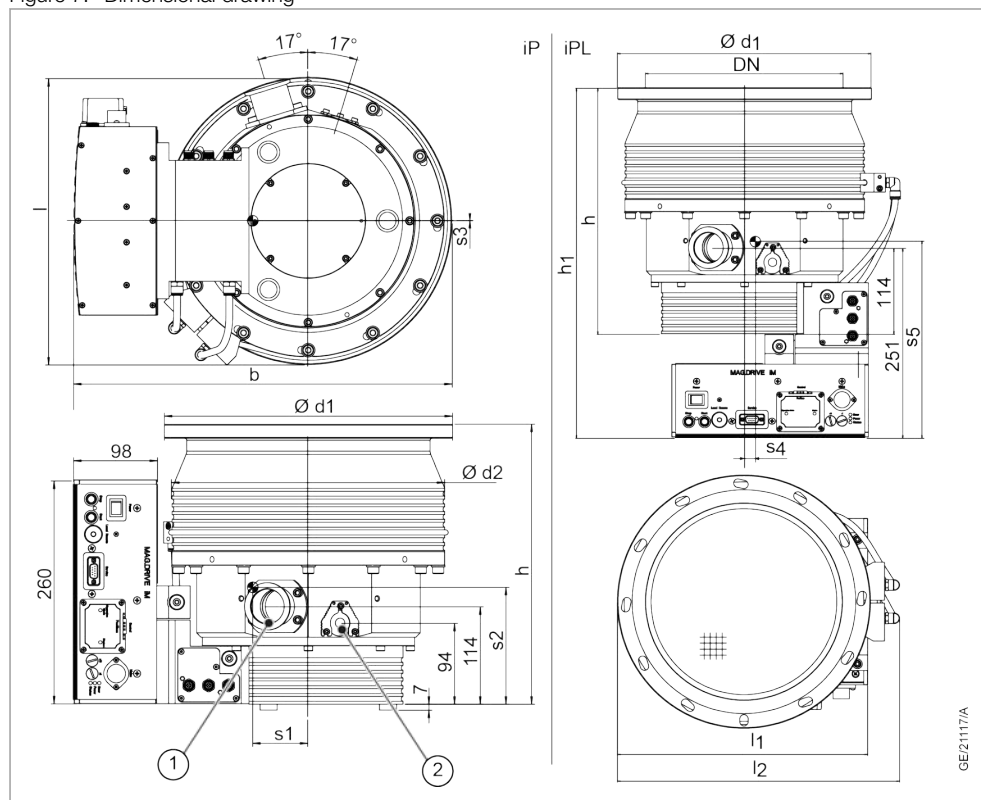


A. Hydrogen
C. Argon

B. Helium
D. Nitrogen

Technical data

Figure 7. Dimensional drawing



MAG	DN	Ød1	Ød2	h	h1	l	l1	b	s1	s2	s3	s4	s5
W1300	200 ISO-F	285	285	305	442	311 *	311 *	332 *	420 *	42	140	0	15
	200 CF	254	285	335	472	307	307	–	416	32	164	0	–
W1600/1600 W1700	200 ISO-F	335	317	325	463	335	331	374	442	39	154	0	14
	200 CF	305	317	330	467	335	322	–	432	29	173	0	9
W2200/2200	200 ISO-F	335	349	355	492	343	340	392	450	34	165	0	12
	200 CF	305	349	372	509	339	340	–	446	26	187	0	8

* 4 mm for cooling coil.
All dimensions in mm.

5 Transportation



WARNING: SUSPENDED LOAD

Risk of injury. Do not stand below the pump while connecting or removing the turbomolecular pump. The pumps are heavy products. Use a crane or suitable lifting equipment to lift the pump.

The devices are delivered in safe transport packaging. Check whether the packaging has been damaged during transport. If that is the case, check the equipment immediately and notify the freight forwarder and us, if necessary.

Keep transport packaging for any further transportation and storage of the pump. Be careful not to damage the sockets and connections during transport.

Unpacking

The turbomolecular pump is shipped in a bag. You must open the bag only when the pump is ready for installation. Do not remove the covers and blanking flanges until you are ready to make the connections, to make sure that the turbomolecular pump is installed under the cleanest possible conditions.

Lifting

The pump is equipped with three eye-bolts which have been screwed into the casing. They can also be screwed into the bottom of the pump. For this, unscrew the feet first. Remove the pump from the packaging using lifting equipment.

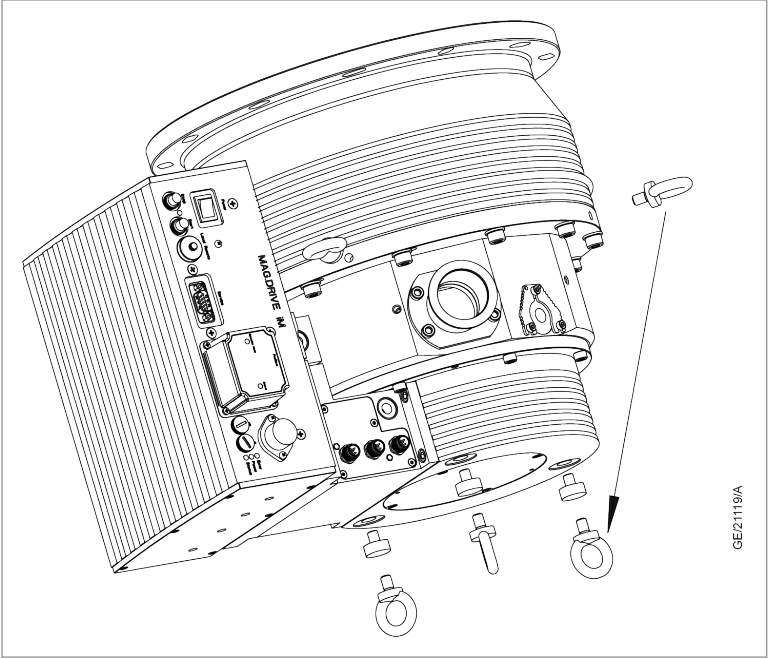


CAUTION: FALLING PARTS

Risk of injury from falling parts. Mistakes made during transport can cause the pump to fall down. Transport the pump only in its transport packaging or by using suitable eye-bolts.

Transportation

Figure 8. Crane eyes



GE2119/A

6 Installation

6.1 Conforming utilization

The pump is intended for generating a vacuum. It is suited for non-corrosive processes only. It must be bolted to a rigid vacuum system and connected to a suitable backing pump.

The pump must only be operated with correspondingly specified frequency converters, the special connecting cables and mounting bolts.

Both pump and frequency converter are intended for being operated within closed rooms.

The use of any accessories which have not been specified by us is only allowed once approved by us.

The supplied integrated MAG.DRIVE iM frequency converter will be required for the operation of the turbomolecular pump.

6.2 Non-conforming utilization



WARNING: NON-CONFORMING UTILISATION OF PUMP

Risk of injury and damage to equipment. Any non-conforming utilisation of pump, frequency converter and accessories can result in severe injury or death and cause damage to components.

Non-conforming utilizations for both pump and frequency converter are among others:

- Pumping of gases and vapours for which the materials of the pump are unsuitable.
- Pumping of toxic gases.
- Operation in connection with processes in which GaAs (gallium arsenide) is being pumped.
- Pumping of gas mixtures with an oxygen content of > 21%.
- Pumping of corrosive gases.
- Pumping of dust containing gases without purge gas operation.
- Pumping of condensable vapours without suitably controlling the temperature of the pump. Upon compression within the pump, these vapours may condense or form deposits.
- Pumping of dusts and solids without the use of suitable screens and filters.
- Operation at an inadmissibly high forevacuum pressure.
- Operation at inadmissibly high gas loads.
- Utilization of both pump and frequency converter in explosion hazard areas.
- Non-compliance of the specified maintenance and servicing intervals for both pump and frequency converter.
- Operation of the pump and drive electronics in environments which demand protection type IP 54 or higher and where the installation site is over 1000 m above sea level.
- Utilization in systems and pump systems in which the pressure may exceed 1.4 bar abs.
- Operation with an inadequately mounted pump.

Installation

- Operation without having flanged the pump to the system or without having connected it to a suitable backing pump.
- Operation with additional heat sources involving thermal radiation, thermal conduction via the high vacuum or the forevacuum flange, strong magnetic fields or very hot process gases, for example.
- Use in systems in which impact stress and vibrations or periodically occurring forces affect pump, frequency converter and cables.
- Operation on moving system or system components (locks or movable pump systems, for example).
- Operation at vibration absorbers and vacuum components (gate valves, valves) which are not capable of sustaining the specified deceleration torque should the pump rotor seize.
- Stepping on pump, add-on parts, drive electronics, flanges and cables to climb onto the system.
- Fitting of add-on parts to the forevacuum flange which cause an inadmissible high load.
- Removing, covering or obstructing warning notices.
- Standstill or storing of pump and drive electronics without suitable sealing off and drying. Storing in a humid atmosphere can cause corrosion.
- Conversions, manipulations and maintenance work by personnel not authorised by us.

6.3 Operating environment

The maximum permissible ambient temperature is 45 °C (113 °F). Do not expose the pump or the frequency converter with type of protection IP 30 to dripping or spraying water.

If the pump is used within a magnetic field, the magnetic induction at the surface of the pump housing may not exceed:

$B = 5 \text{ mT}$ if impinging radially and

$B = 15 \text{ mT}$ if impinging axially.

Exceeding this limit can cause excessive rotor heating due to the eddy currents generated in this situation. It is therefore necessary to provide suitable shielding in such cases.

Places of installation up to 1000 m above sea level (3300 ft) are possible without restrictions. At altitudes over 1000 m heat dissipation by the ambient air is impaired. Please consult us.

The frequency converter must not be operated in explosive gas atmospheres.

6.4 Attach the pump to the vacuumchamber



WARNING: ROTOR DAMAGE

Risk of injury and damage to equipment. Do not touch the rotor. It can cause injury and damage the rotor bearing.



WARNING: EXPLOSIVE MATERIALS

Risk of injury. The high vacuum flange must be firmly mounted to the vacuum chamber. Observe safety information given in [Mechanical hazards](#) on page 10.

The shipping flange and bolts may be used only for shipping purposes. They are not suitable for mounting the pumps in systems.

Remove the transport seal from the intake flange and remove the desiccant. Pay attention to maximum cleanliness when connecting.

Torques when the rotor seizes

In the case of a sudden rotor vane rupture or rotor-stator contact which may occur in practice (caused, for example, by solid objects from the process chamber entering the pump through the high vacuum flange), a crash momentum amounting to 5 kNm maximum needs to be absorbed by the system.

When not complying with the installation and operating conditions or the maintenance intervals described in the Operating Instructions, the rotor may suffer complete destruction by material fatigue, for example. In this case which is highly unlikely to occur in practice, a deceleration or burst momentum needs to be absorbed by the system, should the pump suddenly seize:

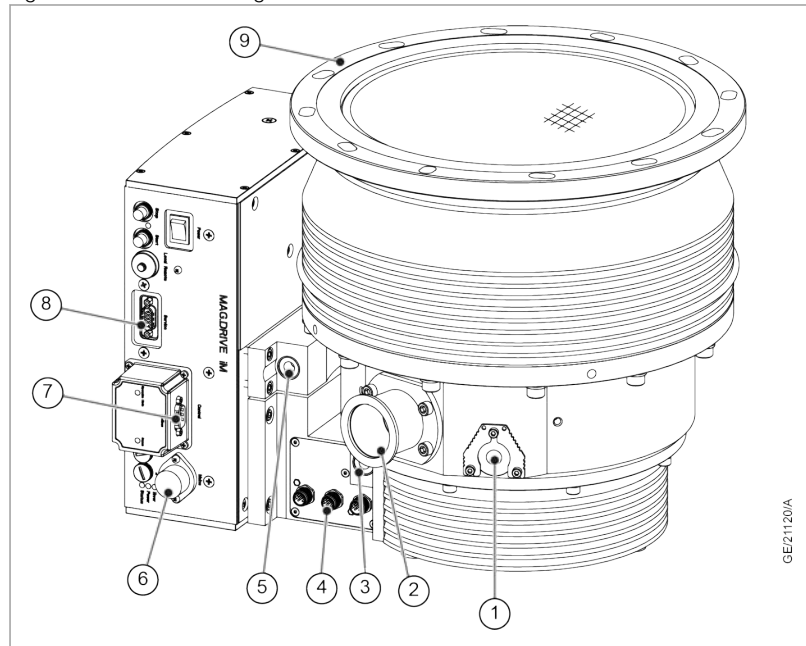
max. 35 kNm	for MAG W 1300 / MAG W 1600 / MAG 1601 / MAG W 1700
-------------	---

max. 50 kNm	for MAG W 2200 / MAG 2201
-------------	---------------------------

We have successfully tested the pumps in agreement with ISO 27892 both against crashes and also bursts.

Installation

Figure 9. Connection fittings



- | | |
|-------------------------------|-----------------------------------|
| 1. Purge gas flange | 2. Forevacuum flange |
| 3. Cooling water OUT | 4. Control connections for valves |
| 5. Cooling water IN | 6. Mains connection |
| 7. Interface (Profibus drawn) | 8. Service interface (RS 232) |
| 9. High-vacuum flange | |

In most applications the pump is flanged to the high-vacuum flange at the apparatus. The pump can be mounted and operated in any desired attitude.

Use exclusively flange connecting components and fittings which have been manufactured in accordance with DIN 28404 (ISO-K flange connections) or ISO 3669 (CF flange connections).

Basically the frequency converter can be detached from the pump. However, this must only be done by an expert trained by Leybold. For this consult with us.

Mount the turbomolecular pump as close as possible to the vacuum chamber. If the turbomolecular pump is permanently flanged to a vacuum chamber with a weight exceeding 500 kg, it will not be necessary to secure it in any other way.

The pump runs low in vibration and noise. No vibrations or resonances from outside equipment may be allowed to be transferred to the pump.

The pump is sensitive to low-frequency vibrations.

The flange material to which the pump is bolted must have at operating temperature a minimum strength specification of 150 N/mm².

Vibration influence

If several turbomolecular pumps are installed to the vacuum chamber of the same system, there is the risk of interference (vibration interference between the pumps). If such a risk exists, contact us for application support.

Earthquake protection

The standard pump fixing is sufficient for earthquake protection. If required fix the system to the bottom or to the walls. Depending on the chamber's weight and fixing use the boreholes in the pump's bottom in addition.

In case of an earthquake (strong vibrations) the pump switches off automatically.

Isolation valve

We recommend installing an isolation valve between the pump and the chamber. The valve should be closed during wet cleans of the chamber and in case of pump failures which will lead to a pump shut down. The valve should normally be closed with power off.

Figure 10. Fixing to the system

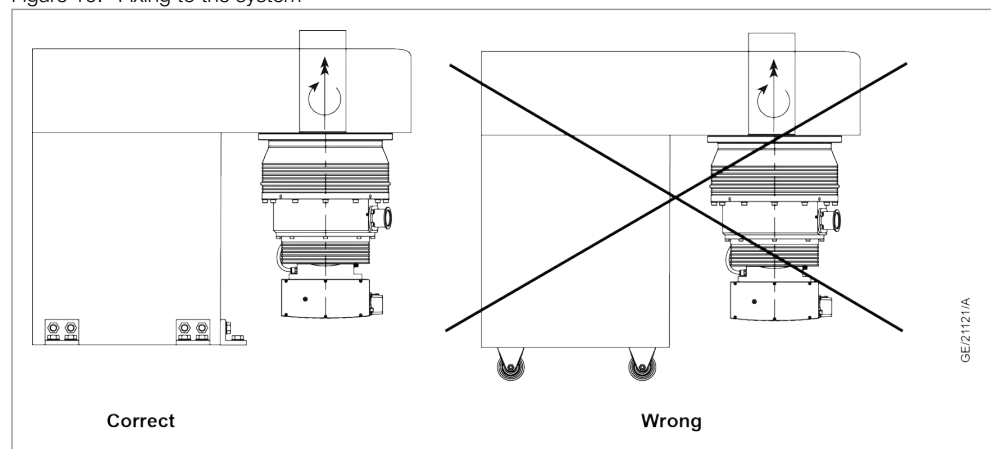
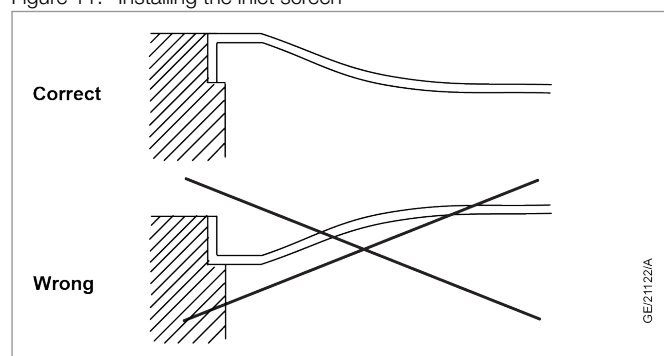


Figure 11. Installing the inlet screen



Install an inlet screen



NOTICE:

Foreign objects which enter the pump through the intake would cause serious damage to the rotor. That's why we recommend installing an inlet screen. Damages caused during operation without the inlet screen are excluded from warranty.

Insert the inlet screen so that the surface curvature is at the bottom and apply some pressure lightly at the rim so that the inlet screen engages, see [Figure: Installing the inlet screen](#) on page 31.

Installation

If dust could pass from the vacuum chamber into the pump, then a micropore filter must be installed between the vacuum chamber and the pump.



NOTICE:

The contact surfaces of pump housing, vacuum system and centering ring must be free of grease and dry so as to ensure adequate strength in case the rotor should seize.

Flange mounting for ISO-F flanges

Mount the turbomolecular pump according to [Figure: Mounting high vacuum flange ISO-F](#) on page 32 and tighten the bolts crosswise step-by-step.

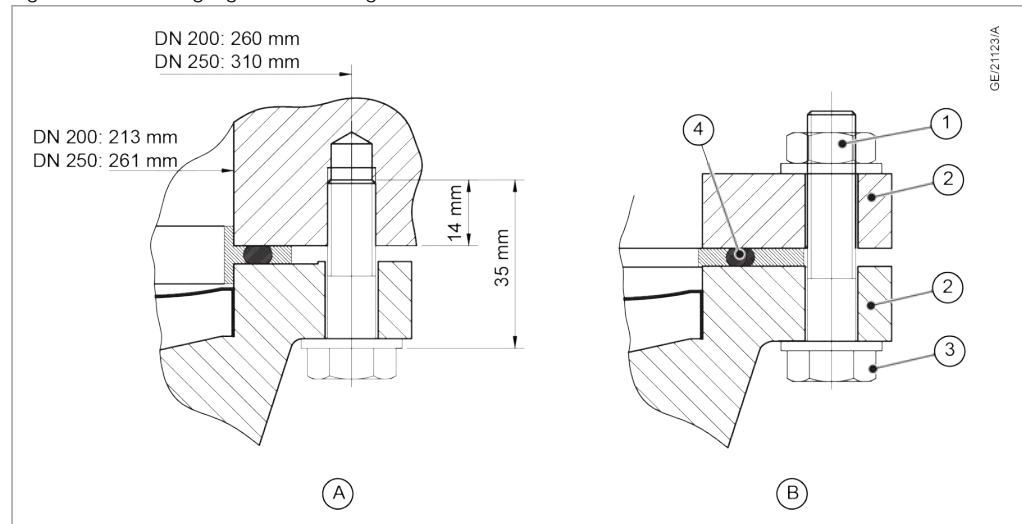
Flange mounting for CF flanges

Before fitting, check to ensure that the sealing edge is undamaged. Do not touch the copper gasket and the sealing edge with your bare hands.

Mount the turbomolecular pump according to [Figure: Mounting high vacuum flange CF](#) on page 33 and tighten the bolts crosswise step-by-step.

When the pump shall be baked out, the threads of the bolts should have been lubricated with a high temperature lubricant.

Figure 12. Mounting high vacuum flange ISO-F



- 1. Nut
- 3. Bolt

- 2. ISO-F flange
- 4. Vacuum sealing disk (centering ring and O-ring with outer support ring)

A. 12 bolts M10 x 35

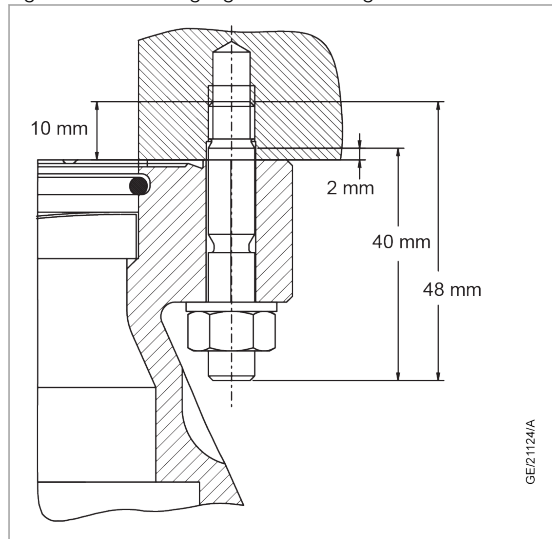
B. DN 200/250 ISO-F: 12 bolts M10 x 50

Installation torque per bolt 35 ⁺⁵ Nm

Bolt quality: 12.9 according to EN ISO 898-1 with coating 0.2% yield strength > 1080 N/mm²

Installation

Figure 13. Mounting high vacuum flange CF



DN 200 CF: 24 bolts M8 x 40

DN 250 CF: 32 bolts M8 x 40

Installation torque per bolt 25^{+5} Nm

Bolt quality 10.9 acc. to EN ISO 898-1 with coating 0.2% yield strength $> 900 \text{ N/mm}^2$

Owing to the deformation of the copper gasket, the fastening torque of all bolts must be checked once more after having completed the installation work.



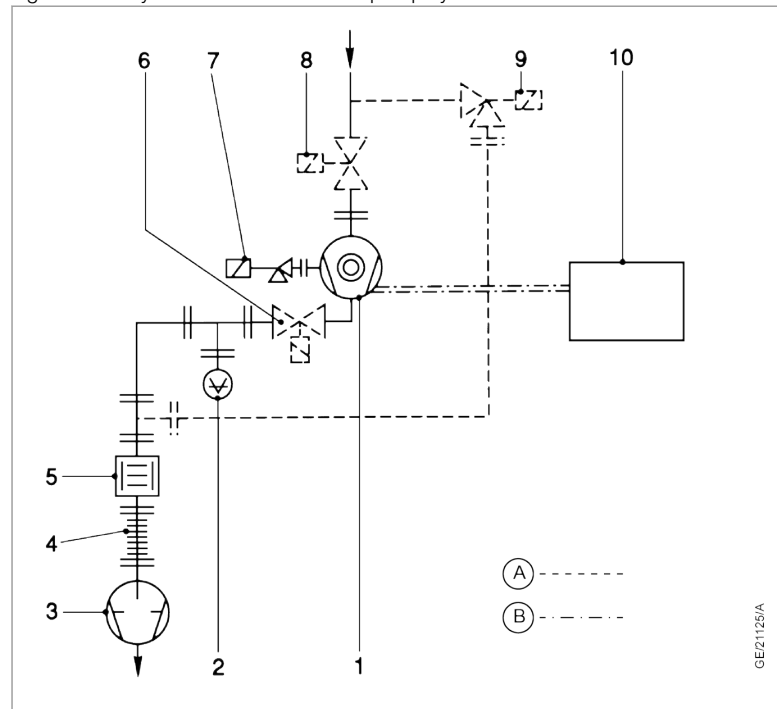
WARNING: HOT SURFACE

Risk of injury. During the operation, the pump can get so hot that there is the risk of suffering burns (up to approximately 120°C). Wear PPE to protect the hot parts against being touched.

Installation

6.5 Forevacuum connection

Figure 14. Layout of a turbomolecular pump system



A. Roughing line; recommended if shorter cycle times are to be achieved

B. Drive and control line

- 1. Turbomolecular pump
- 3. Backing pump
- 5. Sorption trap
- 7. Purge gas and venting valve
- 9. Valve in the roughing line

- 2. Forevacuum gauge point
- 4. Anti-vibration bellows
- 6. Forevacuum valve
- 8. Isolation valve
- 10. Frequency converter



WARNING: TOXIC MATERIAL

Risk of injury. The forevacuum line must be tight. Hazardous gases can escape at leaks or the gases being pumped can react with air or humidity. Observe safety information given in [Hazards caused by materials and substances](#) on page 11.

The high vacuum pressure level which can be achieved is a function of the volume of gas flow Q to be pumped and the forevacuum pressure.

Forevacuum pump

We recommend using our two stage TRIVAC-B pumps or dry compressing pumps as backing pumps.

Connect the clean forevacuum line. The connecting flanges must be clean and undamaged. The cross section of this line must be so wide that safe operation of the pump can be ensured.

[Figure: Layout of a turbomolecular pump system](#) on page 34 shows schematically the design of a pump system incorporating a turbomolecular pump and a TRIVAC backing pump with integral anti-suckback valve.

Safety valve

When using a backing pump not having an integrated anti-suckback valve, a separate safety valve should be used. The safety valve keeps oil from back-streaming from the backing pump and into the turbomolecular pump when the system is not running.

Adsorption trap

To ensure that the forevacuum space at the turbomolecular pump is kept largely free of oil vapors during operation. We also recommend installing an adsorption trap in the forevacuum line. Alternatively purge the forevacuum line with inert gas. In this case the pressure in the forevacuum line must be over 10^{-2} mbar.

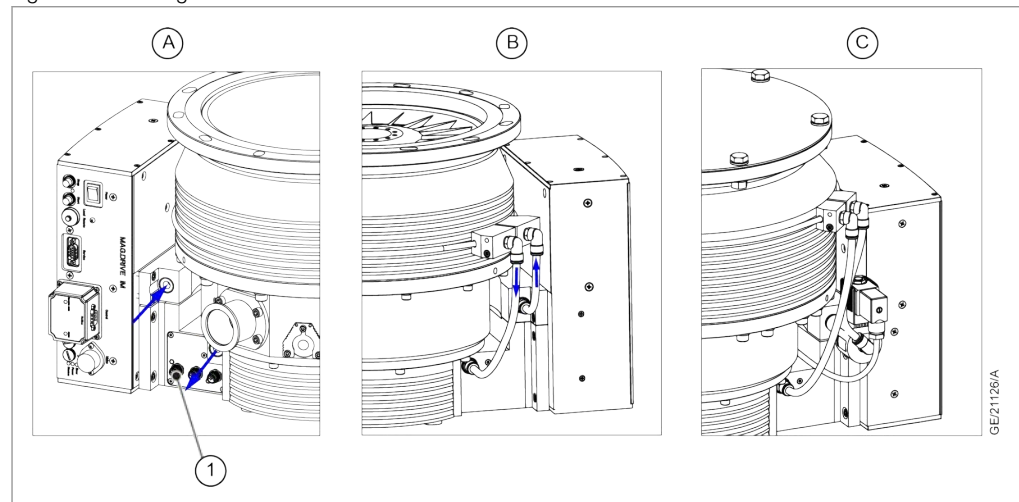
Provide a roughing line to achieve the shortest cycle times.

Ensure that the pump is sufficiently isolated against vibrations generated by the forevacuum pump.

No forces from the piping system may be allowed to affect the turbomolecular pump. Support the piping correspondingly or decouple through flexible joints.

6.6 Connecting the cooling water

Figure 15. Cooling water connection



- A. Cooling water connections
Threads G1/8"
- B. Cooling water connections at the
rear of the pump (without cooling
water valve)
- C. Cooling water connections at the
rear of the pump (with cooling
water valve)
- 1. Electrical connection for the
cooling water valve (optional)

Screw on the cooling water lines. The direction of cooling water is not specified.

Avoid condensate formation

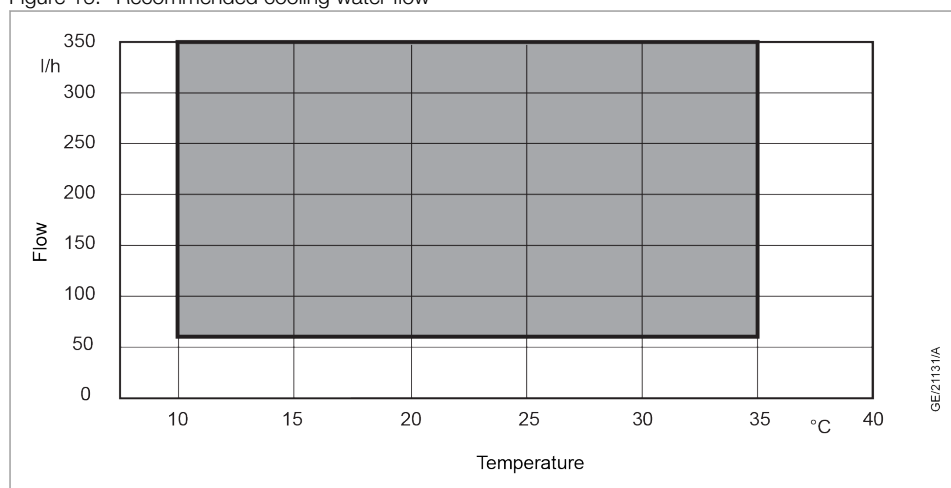
Adjust the cooling water temperature so that the formation of condensate is avoided.

Installation

Turn off the cooling water supply before venting the turbomolecular pump and when it is not running to avoid condensate formation in the pump.

The optional cooling water valve shuts off the cooling water flow when the pump is not turning ($n < 15$ Hz) thereby preventing the formation of condensate within the pump. Install the valve as depicted in [Figure: Cooling water connection](#) on page 35 and electrically connect it to the frequency converter, see fig. 3.11.

Figure 16. Recommended cooling water flow



When using the optional cooling water valve kit the maximum flow is reduced to 150 l/h.

If immediate pump shut-down in case of cooling water supply failure is required, then a flow monitor will have to be inserted in the drain line. If you do not close the cooling water it may take longer to achieve ultimate pressure after start up of the system.

When decommissioning the pump, drain out the cooling water and blow out the cooling water lines so as to avoid frost damage.

Do not open or modify the cooling water connections at the rear of the pump.

Cooling water specifications

Inlet temperature	10 - 35 °C
Inlet pressure	max. 6 bar
Cooling water requirement	> 60 l/h

6.6.1 Water quality

To make sure long trouble-free operation the cooling water must not contain any oils, greases and suspended solids. Moreover, we recommend compliance with the limit values given in [Table: Water quality](#) on page 36.

Table 4. Water quality

Parameter	Value
Appearance	Clear, free of oils and greases
Suspended matter	< 250 mg/l
Particle size	< 150 µm

Installation

Parameter	Value
Electrical conductivity	< 700 µS/cm
pH value	7.0 to 9.0
Total hardness (total alkaline earths)	< 8 °dH
Aggressive carbon dioxide	None, not detectable
Chloride	< 100 mg/l
Sulphate	< 150 mg/l
Nitrate	≤ 50 mg/l
Iron	< 0.2 mg/l
Manganese	< 0.1 mg/l
Ammonium	< 1.0 mg/l
Free chlorine	< 0.2 mg/l



Note:

8 °dH (degrees German hardness) = 1.4 mmol/l

= 10 °e (degrees English hardness)

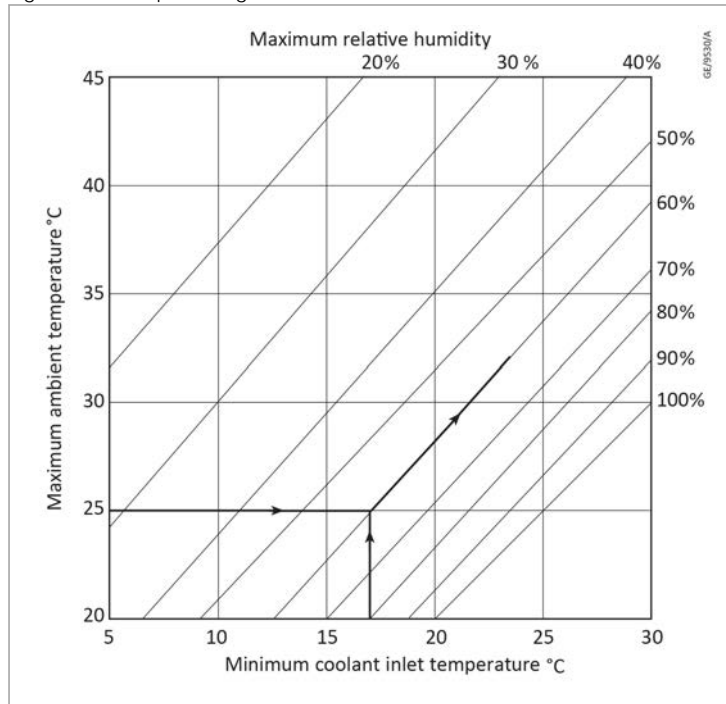
= 14 °f (degrees French hardness)

If there is the danger of frost, you may use a water glycol mixture of up to 30 %.

Deionized water can be used for cooling the pump, if the pH value corresponds to the range indicated above.

Installation

Figure 17. Dewpoint diagram



Minimum coolant inlet temperature at which condensation does not yet occur, as a function of maximum room temperature and maximum relative humidity.

Example:

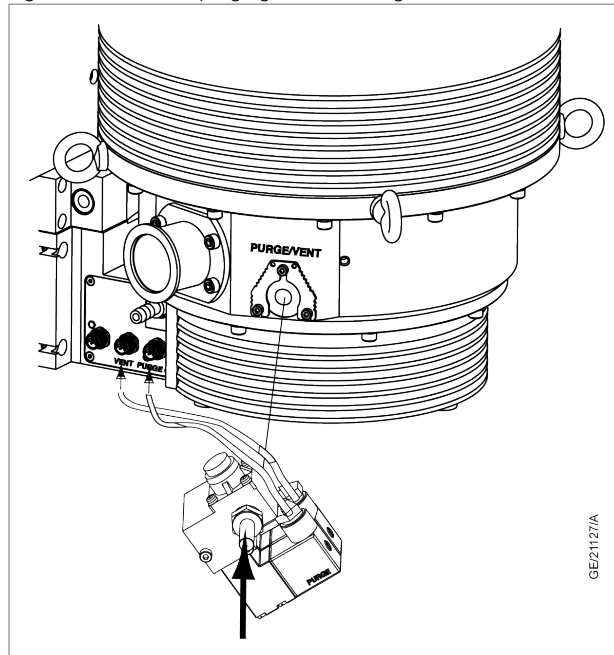
Maximum ambient temperature 25 °C

Minimum coolant inlet temperature 17 °C

⇒ Maximum relative humidity 60%

6.7 Connect the purge gas and venting valve

Figure 18. Connect purge gas and venting valve



WARNING: EXPLOSION HAZARD

Risk of injury or damage to equipment. Make sure that the pressure in the pump must not exceed 1400 mbar (0.4 bar overpressure).

Observe Safety Informations [Important safety information](#) on page 10.

The pumps are equipped with a purge gas facility. A purge gas and venting valve or a power failure venting valve or a venting valve may be connected directly using a DN 16 KF centering ring.

The power failure venting valve or venting valve vents the pump and the forevacuum line when the pump is switched off and thus keeps oil vapor from diffusing back from the forevacuum line.

A choke nozzle in the vent port ensures that the pump is not vented too fast.

When having to decide which gases need or not need to be pumped with purge gas we are available to provide assistance.

Refer to [Media compatibility / purge gas](#) on page 45 for compatible gases.

When operating the pump with purge gas, the pump needs to be vented via the purge gas valve after having shut down the pump, see Section [Venting](#) on page 48.

Consider the additional purge gas flow when selecting a suitable backing pump.

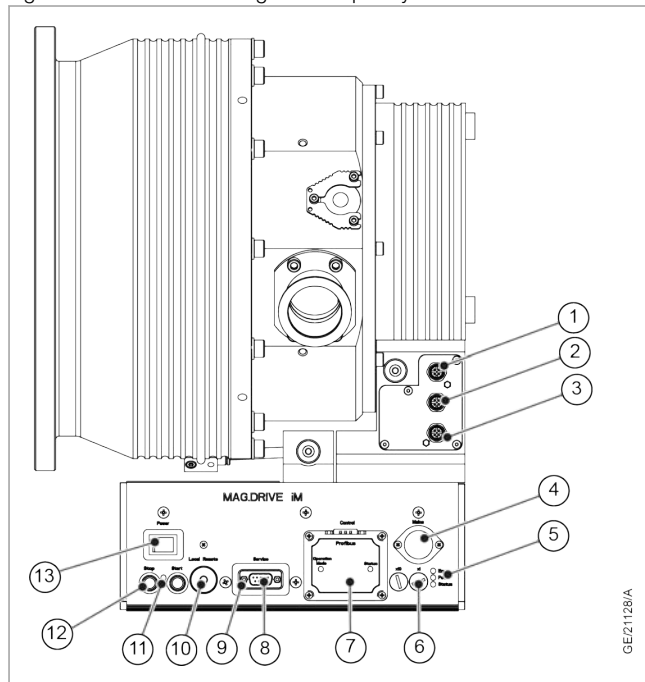
We recommend a purge gas flow of 0.6 mbar·l/s (36 sccm) with Nitrogen.

Connect the purge gas and the venting valve recommended by us according to its instructions. The valve cables can be connected directly at the pump.

Installation

When doing so, do not confuse “Purge” and “Vent”!

Figure 19. View of the integrated frequency converter



- | | |
|--|--|
| 1. Purge gas | 2. Venting |
| 3. Cooling water | 4. Mains cable |
| 5. LEDs | 6. Address switches |
| 7. Interface (Profibus or optional) | 8. Service interface (RS 232) |
| 9. UNC 4-40 threads | 10. Switch Local/Remote, (under a cover) |
| 11. LED Local/Remote, lights at Local | 12. Start/Stop keys |
| 13. Main switch (do not use to stop the pump!) | |

6.8 Electrical connection for the integrated frequency converter



WARNING: HAZARDOUS VOLTAGES

Risk of electric shock. Observe Safety Informations in [Electrical hazards](#) on page 10.



WARNING: HOT SURFACE

Risk of injury. During operation, the frequency converter may attain temperatures up to 75 °C. We recommend that the unit be installed so that it can not be touched inadvertently.



NOTICE:

Disconnect and connect the cable connections only while the pump is turning no longer (green status LED off) and with the mains power switched off (yellow power LED off). Otherwise there is the risk of damaging the frequency converter.

The interface connectors have UNC 4-40 threads. Do not use connectors with M3 threads.

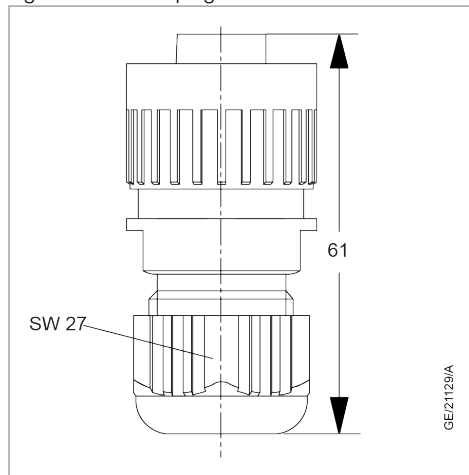
The pump may be operated only with suitable connector cables.

Route all cables so, as to protect them from damage.

Plug in the mains cable. Refer to Fig. 3.12 for the mains plug connection.

In order to comply with the standard SEMI S2, install an additional mains power circuit breaker at the sides of the system. The short-circuit switch-off capacity of this mains power circuit breaker needs to be rated at 10,000 A minimum.

Figure 20. Mains plug



Connections:

- 1 Phase L1
- 2 Neutral N
- 3 not connected
- PE Protective earth

Wire cross section max. 2.5 mm²

Cable diameter 6-12 mm

Installation

6.9 Relays, LEDs, PLC interface

Table 5 Relay status

Input data / status				Output data					Operating mode
Start/ Stop signal	Pump rotating	Normal fre- quency $\geq$ 90% of set- point fre- quency	Error is present	Motor drive	Relay Normal operation	Relay error	LED sta- tus	LED error	Other modes are not possible; they indi- cate a failure affecting the TURBO.DRIVE.
Stop	no	no	no	off	passive	passive	off	off	Pump not operating
Stop	yes	no	no	off	passive	passive	flashes	off	Pump is decelerating
Stop	yes	yes	no	off	passive	passive	flashes	off	Just after stop; pump was in the normal oper- ating mode before that
Start	no	no	no	on	passive	passive	off	off	Just after start
Start	yes	no	no	on	passive	passive	flashes	off	Pump is accelerating
Start	yes	yes	no	on	active	passive	green	off	Pump is in the normal operating mode
Stop	no	no	yes	off	passive	active	off	red	Error is present; pump is at standstill
Stop	yes	no	yes	off	passive	active	flashes	red	Error is present; pump is decelerating
Stop	yes	yes	yes	off	passive	active	flashes	red	Error has just occurred
Start	no	no	yes	off	passive	active	off	red	Error is present; pump is at standstill
Start	yes	no	yes	off	passive	active	flashes	red	Error is present; pump is decelerating
Start	yes	yes	yes	off	passive	active	flashes	red	Error has just occurred

6.9.1 PLC interface

Figure 21. Pin assignment of the connector

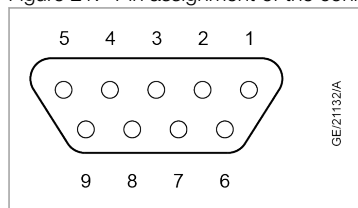
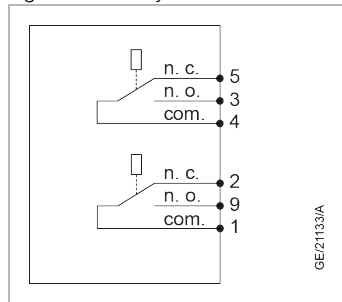


Figure 22. Relay functions



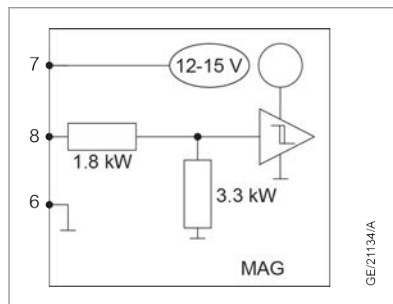
Only for SELV or PELV power circuits up to 32 V, maximum current 0.5 A

Relay - Normal operation (standard setting)

- While deceleration, acceleration, Stop: 4 connected to 5 (as shown; passive)
- During normal operation ($f > 0,9 \cdot f_{nom.}$): 4 connected to 3 (active)

Relay - Error

- No error: 1 connected to 2 (as shown; passive)
- Error is present: 1 connected to 9 (active)



Switching threshold for the Start/Stop control input:

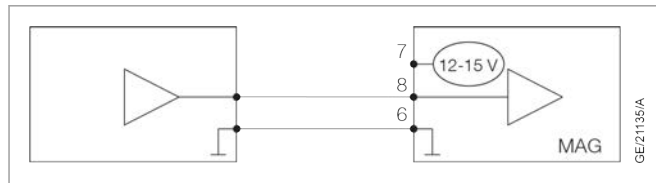
Low level: $< 8 \text{ V}$

High level: $> 10.5 \text{ V}$

Installation

Start /stop function

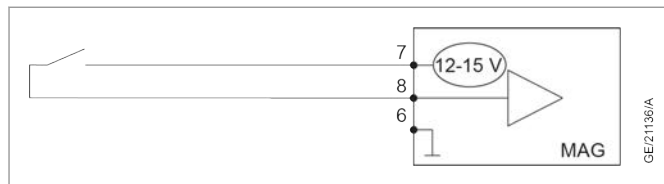
Example 1: Operation via a PLC



0 V = STOP

12-24 V = START

Example 2: Operation via contacts



Contact open = STOP

Contact closed = START

6.9.2 Front LEDs

ERR ○ PWR ○ STS ○	Red LED ERR	off	No error, no warning
		flashes	Warning is present, pump can be operated, possibly with some restrictions
		on	Fault is present, pump stopped or can not be operated
	Yellow LED PWR	off	No supply voltage
		flashes	Supply voltage too low or too high, 9 Hz < n < 100Hz (venting possible)
		on	Supply voltage is present
	Green LED STS	off	Pump at standstill (< 9 Hz)
		flashes slowly 1/s	Start command is present
		flashes fast 3/s	Running down, brake operation
		flashes shortly (0.3 s)	Start delay active
		on	Normal operation
	Yellow LED PWR and green LED STS	Flashes fast alternately	Running down, brake operation at n < 100 Hz i.e. venting is possible.

7 Operation

7.1 Media compatibility / purge gas

The MAG turbomolecular pumps are suitable for pumping air and clean gases.

The MAG ... iP model include a purge gas device.

We are available for consultation regarding the media that can be safely handled with this unit.

Install a micropore filter when pumping media which contains dust.

Compatible gases

All gases suitable for venting or purging,

- must not cause corrosion or pitting in aluminum and steel
- should not cause corrosion or sticking when interacting with process deposits in the pump.

For venting and purging, we recommend using inert gases such as nitrogen or argon. The temperature of these gases should be maintained between 5 °C and 80 °C, with a maximum relative humidity not exceeding 10 ppm.

The gas must be clean. In specific cases and after consultation, dry, filtered, oil-free air or filtered ambient air (filter mesh < 1µm) may also be used. Replace the filters periodically, at least once a year.

7.2 Interfaces

The MAG.DRIVE can optionally be equipped with the following interfaces:

- RS 232
- RS 485
- Profibus
- PLC (9-pin Sub-D)
- Ethernet/IP (1-port)
- EtherCat (in preparation)

Type of protection

If the interface modules are not covered, they will comply with type of protection IP 30, with cover to IP 54. The remainder of the pump and the frequency converter will in the case of properly connected cables always comply with IP 54.

If you try to control via two installed interfaces, the interface in the CONTROL slot will have the higher priority.

The address for the Fieldbus module can be set via the RS 232 service interface or at the address switches.

The set default address is 126dec = 7Ehex.

Operation

7.3 Switching ON



NOTICE:

Do not open the turbomolecular pump suddenly to a previously evacuated vacuum chamber or to a large-volume forevacuum line which has already been evacuated. The pressure surge can press against the rotor into the limiting bearing, causing accelerated wear at that bearing.

Turn on the backing pump. Activate the main switch; the Power ON LED will light up green.

During the initialization phase (approximately 45 seconds), all LEDs on the frequency converter will alternate, followed by the yellow LED lighting up.

Start the turbomolecular pump by:

- pressing the START button (in LOCAL mode) or
- via the interface, e.g., Profibus.

The turbomolecular pump will begin to run, indicated by the flashing green LED. Once the pump reaches normal operation, the green LED will remain lit. The backing pump and the MAG can be activated simultaneously, allowing the turbomolecular pump to act as an effective baffle from the start.

If the turbomolecular pump is to be activated after a delay, pre-evacuation can occur through the turbomolecular pump even when it is not running.

7.4 Operation

Protecting the touch-down bearings

The magnetic bearing in the MAG are immune to wear. In addition to the magnetic bearings, the MAG is equipped with touch-down bearings which protect the rotor against mechanical contact with the stator if the pump is subjected to external shock loading or when the pump is switched off. These touch-down bearings have a limited service life. Please observe the following in order to obtain maximum service life.

- Avoid shock and vibrations (e.g. from other pumps) when the pump is running. Shocks are particularly harmful. If the pump appears to be running in the mechanical bearings continuously it is switched off.
- Avoid a frequent switching on and off.
- Do not suddenly expose the MAG to an already evacuated vacuum chamber. The pressure surge may cause the rotor to make contact with the touch-down bearings. This will cause increased wear.
- Do not stop the Mag with the mains. Use a stop command. Switching off the mains while the pump is running will wear out the touch down bearings. If the mains supply has been disconnected accidentally re-connect it.

7.5 Switching off

CAUTION: AUTOMATIC START UP



Risk of damage to equipment. After a mains power failure the pump can run up automatically. This is intended to keep the vacuum during short mains failures.

Also, at speeds approximately below 100 Hz there will not be enough power any more for the LEDs, i.e. the pump may still run with out any of the LEDs being ON. The operator must ensure safety by applying suitable measures.

CAUTION: ELECTRICITY HAZARD



Risk of injury. Unplug any connectors only when the mains voltage is switched off and the pump no longer turns.

NOTICE:



Do not stop the Mag with the mains. Switching off the mains while the pump is running will wear out the touch down bearings.

Switch off the pump

- by pressing the STOP button (in the LOCAL mode)
- via the interface, e.g. Profibus.

After switching off, the green status LED will flash until the rotor of the turbomolecular pump is at standstill. This may take several minutes. The deceleration time of the pump corresponds approximately to the run-up time of the pump. The rotor may be stopped faster by venting the pump. The pump must only be handled with the rotor not rotating.

With the DC power supply off, the turbomolecular pump will act as a generator supplying the frequency converter with energy as indicated by the yellow power LED.

Switch off the forevacuum pump.

Venting

When using oil-sealed forevacuum pumps, vent the turbomolecular pump before it comes to a stop; refer to Section [Venting](#) on page 48.

When using TRIVAC pumps the built-in anti-suckback valve will close automatically, shutting off the forevacuum line. In forevacuum pumps without a vacuum retention valve, close the valve in the forevacuum line.

When the system is not operating, ensure that neither ambient air nor cleaning media can enter the pump.

If a failure occurs the turbomolecular pump will be shut down automatically.

The red LED at the frequency converter lights up.

Emergency shut down

In the case of an emergency shut down, the pump is switched off as described above. The rotor of the turbomolecular pump may be stopped

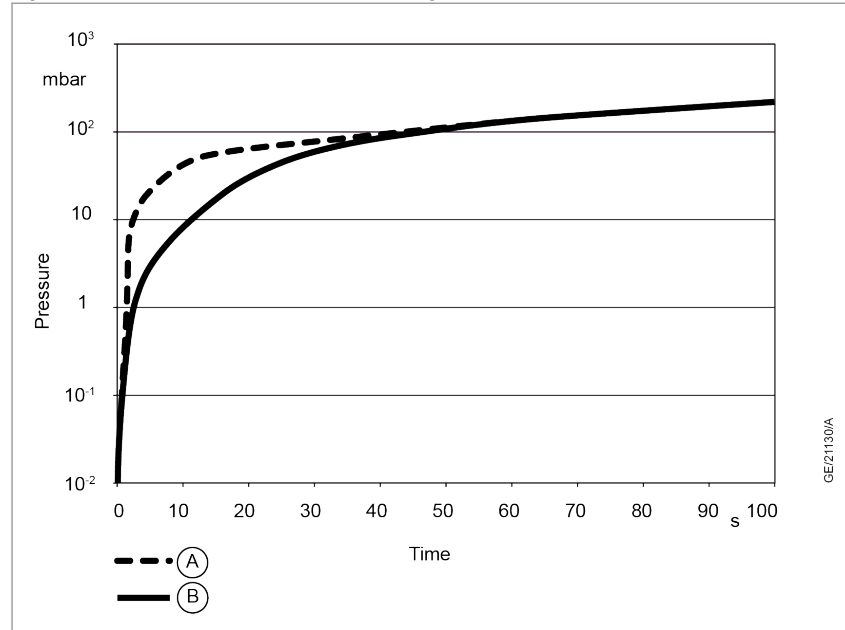
Operation

faster by controlled venting the pump, see [Figure: Pressure rise curve for safe venting](#) on page 48.

7.6 Venting

Vent the turbomolecular pump each time it is shut down, in order to prevent any return diffusion of oil vapors out of the forevacuum line and into the highvacuum side.

Figure 23. Pressure rise curve for safe venting



A. Max. forevacuum pressure

B. Max. high-vacuum pressure

Venting methods

There are three different methods of venting the turbomolecular pump.

In the case processes requiring a purge gas, the pump must be vented via the purge gas and venting valve when shutting the pump down.

When additionally venting the vacuum chamber, the venting function of the purge gas and venting valve must be opened before opening the chamber valve. This will ensure the presence of a higher pressure in the area of the bearings compared to the remaining vacuum area. This will prevent particles, dust or aggressive gases from being forced through the bearings into the not yet vented motor chamber of the pump.

Cautious venting of the pump is possible from the high vacuum side, since here the bearing forces will be lowest. When doing so, no free jet of gas must be allowed to form on the rotor so as to avoid exposing the rotor to additional forces.

When venting the pump through its foreline connection, neither oil nor particles may be entrained in the gas flow from the forevacuum side into the pump.

Speed of the Pressure Rise

All turbomolecular pumps may be vented at full speed. However, the pressure must not increase faster than specified through the pressure rise curve, see [Figure: Pressure rise curve for safe venting](#) on page 48.

The pump must be vented significantly slower when there is the risk of particles entering into the pump from the process. During venting, the flow must be of the laminar type in both the vacuum chamber and the turbomolecular pump.

The speed of the pressure rise during venting of the running pump will greatly influence the load on the rotor/stator pack and the bearings. The slower the pump is vented, the longer the service life of the bearings will be.

The pump must not be vented to pressures above atmospheric pressure.

7.7 Removing the pump from the system



WARNING: HAZARDOUS GASES

Risk of injury. If the pump has previously handled hazardous gases, implement the proper precautionary measures before opening the intake or exhaust connection.

Observe safety information given in [Hazards caused by materials and substances](#) on page 11.

Shut down the pump and vent as described in [Switching off](#) on page 47 and [Venting](#) on page 48.

Disconnect the pump only when it has completely stopped. The green LED must have gone out.

Then switch the mains power off and wait until the yellow power LED is off.

Then disconnect any cable connections.

Drain cooling water

Drain out the cooling water and blow out the cooling water lines to prevent frost damage.

Hazardous gases/Deposits

The pumps may be contaminated with process gases. These gases may be toxic and hazardous to health. In addition, deposits with similarly dangerous properties may have formed. Many of these gases and deposits form acids when they come into contact with humid air. This will result in serious corrosion damage to the pump.

Desiccant

To avoid health hazards and corrosion when the pumps are detached from the system, fasten a container of desiccant under the transport cover of the high vacuum connection and then close the pump immediately at all flange connections. Store the pump, with a desiccant, in an airtight PE bag.

Corrosion due to faulty packing will nullify the guarantee.

Pack the pump to prevent damage during shipping and storage. Pay particular attention to protection for the flanges and the electrical plug.

Observe the instructions in [Service](#) on page 55 if you forward the pump to us.

8 Maintenance

Rotor exchange

Observe Safety information in [Important safety information](#) on page 10.

MAG W 1300, W 1600, 1601, W 1700: We recommend to change the rotor after 20,000 cycles at the latest.

MAG W 2200, 2201: The rotor must be changed after 10,000 cycles at the latest.

All pumps: We recommend to change the rotor after 80,000 hours of operation at the latest.

Such maintenance work can only be done by us. If required contact our service center nearest to your location. You can find the address on our internet page www.leybold.com.

At high pump loads - for example during cyclic operation, at high gas throughputs or at high ambient temperatures - the aforementioned maintenance work should be carried forward. Please consult us for recommendations.

Purge gas filter

Depending on the degree of contamination of the purge gas used the filter will clog and will have to be exchanged (our experience indicates that this will become necessary after 1 to 6 months).

Adsorption trap

When an adsorption trap is used, regenerate or renew the adsorption agent regularly; refer to the operating instructions provided with the trap.

8.1 Cleaning

If required clean the turbomolecular pump of dust with a dry cloth.

8.2 Changing the touch-down bearings

Wear occurs at the touch-down bearings when hard shocks have to be supported.

The frequency converter outputs a warning and then an error message if the touch-down bearings have had contact too often or for a too long time (default 1000 contacts or 1 hour). In this case maintenance is required.

Only the Leybold service can change the touch-down bearings.

9 Fault finding



WARNING: HAZARDOUS VOLTAGES

Risk of electric shock. When the connector cable is attached, the outputs at the frequency converter are not free of voltage.

Before you start searching for the source of the problem, you should carry out a few simple checks:

Are the electric connections in good working order?

Is the forevacuum pressure sufficient?

After having removed the cause for the error reset the error message at the frequency converter by applying a **STOP** signal via the socket **CONTROL** or by switching the mains power off.

Malfunction	Possible cause	Corrective action	Responsible
Red ERROR LED is on	Forevacuum or high-vacuum pressure too high	Check the forevacuum pump and use a different forevacuum pump if necessary.	Operator/ maintenance staff
	Gas volume too great	Seal leak; install a higher-capacity vacuum pump if necessary.	Maintenance staff
	Bearing defective	Repair the pump.	Leybold service
	Ambient temperature too high	Install additional water cooling or reduce throughput.	Maintenance staff
	Frequency converter faulty	Replace frequency converter.	Maintenance staff/ Leybold service
	EMC influence	Switch the power supply voltage off and then on again.	Operator
Yellow power LED is not on.	No power supply	Check cables and power supply.	Operator/ maintenance staff
	Frequency converter defective	Replace frequency converter.	Maintenance staff/ Leybold service
Red LED flashes.	Warning message. Possible causes as in "Red ERROR LED is on"	The pump can continue to run, as long as operation limits are only exceeded for a short time.	Operator
Yellow LED flashes.	Warning message: Supply voltage too low or too high	The pump can continue to run. Check the reason for wrong voltage and eliminate the fault.	Operator/main- tenance
Turbomolecular pump does not start, ERROR LED does not light.	CONTROL connector connected wrongly	Connect as shown in Figure: Pin assignment of the connector on page 42.	Operator/Main- tenance staff
	CONTROL and SERVICE connectors mixed up	Connect correctly.	Maintenance staff

Fault finding

Malfunction	Possible cause	Corrective action	Responsible
Turbomolecular pump produces loud running noises and vibrations.	Rotor out of balance	Balance the rotor.	Leybold service
	Bearing defective	Replace the bearing.	Leybold service
	Exciting of vibrations by the process	Modify the system.	Maintenance staff
Turbomolecular pump does not reach ultimate pressure.	Measurement instrument defective	Inspect the measurement sensor.	Operator/Main-tenance staff
	Measurement sensors soiled	Clean or replace the sensors.	Maintenance staff
	Leaks at the equipment, lines or the pump	Check for leaks.	Maintenance staff
	Pump soiled	Clean the pump.	Leybold service
	Backing pump provides insufficient pumping speed or ultimate pressure which is too high	Check the ultimate pressure of the backing pump and install a higher capacity vacuum pump if necessary.	Operator/main-tenance staff
	Frequency parameters programmed wrongly	Check parameters.	Leybold service

10 Storage

To store the devices, place them in a sealable PE bag containing a drying agent and use the supplied transport packaging. Do not store the pump and accessories in a moist atmosphere to prevent it from corrosion. Take note of details in [Removing the pump from the system](#) on page 49.

11 Disposal



WARNING: CONTAMINATED PARTS

Risk of injury. Contaminated parts can be detrimental to health and the environment. Before beginning with any work, find out whether any parts are contaminated. Adhere to the relevant regulations and take the necessary precautions when handling contaminated parts.

Waste disposal

The equipment may have been contaminated by the process or by environmental influences. In this case, the equipment must be decontaminated in accordance with the relevant regulations. We offer this service at fixed prices.

Separate clean components according to their materials, and dispose of these accordingly. We offer this service. Further details are available on request.

When sending us any equipment, observe the regulations given in [Service](#) on page 55.

12 Service

12.1 Return the equipment or components for service

Before you send your equipment to us for service or for any other reason, you must complete a Declaration of Contamination Form. The form tells us if any substances found in the equipment are hazardous, which is important for the safety of our employees and all other people involved in the service of your equipment. The hazard information also lets us select the correct procedures to service your equipment.

If you are returning equipment note the following:

- If the equipment is configured to suit the application, make a record of the configuration before returning it. All replacement equipment will be supplied with default factory settings.
- Do not return equipment with accessories fitted. Remove all accessories and retain them for future use.
- The instruction in the returns procedure to drain all fluids does not apply to the lubricant in pump oil reservoirs.

Download the latest documents from leybold.com/en/downloads/download-documents/declaration-of-contamination/, follow the procedure in HS1, fill in the electronic HS2 form, print it, sign it, and return the signed copy to us.



NOTICE:

If we do not receive a completed form, your equipment cannot be serviced.

Accessories

13 Accessories

Table 6. Required accessories

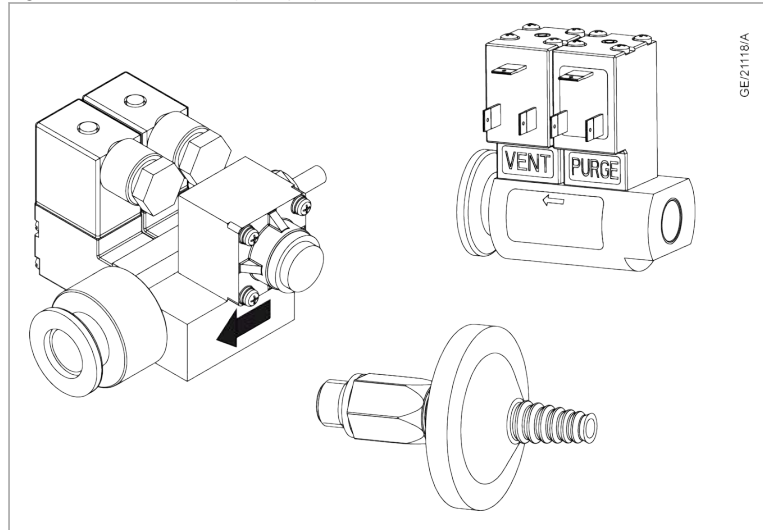
Set of bolts, nuts and washers for ISO-F flange (12 each) Bolts M 10 x 50 Bolts M 10 x 35 Bolt quality 12.9 acc. to EN ISO 898-1 with coating; 0.2 % yield strength > 1080 N/mm ²	400153V0012 400153V0010
Centering ring with O-ring Al/FPM 268 44 268 45 Stainless steel/FPM	DN 200 DN 250 268 44 268 45 887 02 887 08
Set of bolts, nuts and washers for CF flange (8 each) Bolts M 8 x 40 Bolt quality 10.9 acc. to EN ISO 898-1 with coating; 0.2 % yield strength > 900 N/mm ² (for DN 200 3 sets required, for DN 250 4 sets)	400153V0016
Copper gasket rings for CF flange DN 200 (Set of 10 pieces) DN 250 (Set of 5 pieces)	839 47 839 48
Set of hex. screws with nuts, screws and washers for CF flange DN 200 DN 250 (2 sets required)	839 07 839 07

Table 7. Optional accessories

Mains cable for MAG.DRIVE iM, 2.5 m EU US	411310V03 411320V03
Inlet screen DN 200 ISO-F and DN 200 CF DN 250 ISO-F and DN 250 CF	E 200 04 558 E 200 04 557
Seal Kit DN 250 Metal	200 07 901
Purge gas and venting valve 24 V DC 0.6 mbar·l/s at 1.5 to 6 bar 0.6 mbar·l/s at 1 to 1.5 bar Cable set (2 pieces) for connection to the pump	121 33 800152V0010 411300V01
Cooling water valve kit	411300V02

Accessories

Figure 24. Accessories (example)



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EC Declaration of Incorporation

(Translation of original Declaration of Incorporation)

The manufacturer: Leybold GmbH
Bonner Strasse 498
D-50968 Köln
Germany

herewith declares that the following product:

Product designation: magnetic levitated turbo-molecular pump with integrated controller

Type designation:	Part number:
MAG W 1300iP(L)	411300Vxxxx
MAG W 1600iP(L)	411600Vxxxx
MAG W 1700iP(L)	411700Vxxxx
MAG W 2200iP(L)	412200Vxxxx

x=0 until 9

complies with the following fundamental requirements of the **Machinery Directive (2006/42/EC)**:
Annex I, Paragraph 1.1.2, 1.1.3, 1.1.5, 1.2.1, 1.2.3, 1.2.4.1, 1.2.4.2, 1.2.6, 1.3.1, 1.3.2, 1.3.3, 1.3.4, 1.3.7, 1.5.1, 1.5.2, 1.5.4, 1.5.5, 1.5.13, 1.6.1 and 1.7.1

The safety objectives of the Low Voltage Directive 2014/35/EU were complied with in accordance with Appendix 1 No. 1.5.1 of Machinery Directive 2006/42/EC.

The following harmonised standards have been applied:

EN 1012-2:1996+A1:2009	Compressors and vacuum pumps - Safety requirements Part 2: Vacuum pumps
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EN 61010-1:2010	Safety requirements for electrical equipment for measurement, control, and laboratory use — Part 1: General requirements
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The incomplete machine may only be put into operation after it has been determined that the machine into which the incomplete machine shall be installed complies with the regulations laid down in the EC Machinery Directive (2006/42/EG).

The manufacturer commits himself to make the special documentation on the incomplete machine electronically available to national authorities upon request.

The special engineering documentation belonging to the machine was compiled in accordance with Annex VII Part B.

Documentation officer:	Herbert Etges T: +49(0)221 347 0 F: +49(0)221 347 1250 documentation@leybold.com
------------------------	---

Cologne, October 11, 2016

Cologne, October 11, 2016



ppa. Martin Tollner
Head of Product Lines



ppa. Dr. Monika Mattern-Klosson
Head of Quality & Business Process Management

Document No.: 300353440_002_A1

EU Declaration of Conformity

(Translation of original Declaration of Conformity)

The manufacturer:
Leybold GmbH
Bonner Strasse 498
D-50968 Köln
Germany

herewith declares that the products specified and listed below which we have placed on the market, comply with the applicable EU Directives. This declaration becomes invalid if modifications are made to the product without agreement of Leybold GmbH.

Product designation: magnetic levitated turbo-molecular pump with integrated controller

Type designation:	Part number:
MAG W 1300iP(L)	411300Vxxx
MAG W 1600iP(L)	411600Vxxx
MAG W 1700iP(L)	411700Vxxx
MAG W 2200iP(L)	412200Vxxx

x=0 until 9

The products complies to the following Directives:

Electromagnetic Compatibility (2014/30/EU)

The following harmonized standards have been applied:

EN 61326-1:2013	Electrical equipment for measurement, control and laboratory use — EMC requirements — Part 1: General requirements Emissions: Group 1, Class A Immunity: Industrial electromagnetic environment
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Documentation officer:	Herbert Etges T: +49(0)221 347 0 F: +49(0)221 347 1250 documentation@leybold.com
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Cologne, October 11, 2016

Cologne, October 11, 2016



ppa. Martin Tollner
Head of Product Lines



ppa. Dr. Monika Mattern-Klosson
Head of Quality & Business Process Management

Document No.: 300353440_002_A1

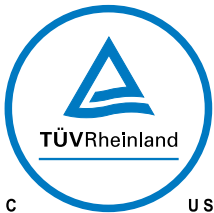
The MAG integra have been tested by the TÜV Rheinland of North America according to the requirements of

- UL 61010-1:2012
- CAN/CSA-C22.2 NO. 61010-1-12

The components are in compliance to the tested standards.

Test Report No. 31782273.001
Certificate No. CU72171775.01

The TÜV Rheinland of North America is a “Nationally Recognized Testing Laboratory” (NRTL) for the USA and Canada.



This product has been tested to the requirements of CAN/CSA-C22.2 No. 61010-1, third edition, including Amendment 1, or a later version of the same standard incorporating the same level of testing requirements.

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Pioneering products. Passionately applied.

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